User Manual

DEV 2142, DEV 2142/N, DEV 2143, DEV 2145
Managed Distribution Amplifier; 800...2450 MHz

DEV 2151, DEV 2151/N
Redundant Amplifier; 800...2450 MHz

DEV 2152, DEV 2152/N, DEV 2153, DEV 2155
Redundant Distribution Amplifier; 800...2450 MHz

DEV 2242, DEV 2242/N, DEV 2243, DEV 2245
Managed Combining Amplifier; 800...2450 MHz

DEV 2252, DEV 2252/N, DEV 2253, DEV 2255
Redundant Combining Amplifier; 800...2450 MHz

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1 Revision History

Revision (Date)	Author	Short Description
A (14-Nov-2019)	WP	Initial version
B (10-Dec-2019)	WP	Additional SNMP OIDs
C (27-Apr-2022)	WP	Additional Button (Switch back to Main)
D (19-Jul-2022)	WP	Minor changes
E (15-Feb-2023)	WP	Added Amplifier and N-Connector Products

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2 Introduction

Thank you for purchasing a DEV 2x4x/DEV 2x5x Distribution/Combining Amplifier. This user manual is intended to familiarize you with the installation and operation of the equipment.

The DEV 214x/DEV 215x series products are single or dual 1:8 or single 1:16 L-Band distribution amplifiers, the DEV 224x/DEV 225x series products are single or dual 8:1 or single 16:1 L-Band combining amplifiers, and The DEV 2151 is a single or dual redundant L-Band amplifier i.e., one input port and one output port per amplifier. In addition, a few products are offered, that provide 50 Ohm female N-connectors: DEV 2151/N (single or dual L-Band amplifier), DEV 2142/N & DEV 2152/N (single 1:8 L-Band distribution amplifiers), and DEV 2242/N & DEV 2252/N (single 8:1 L-Band combining amplifiers).

The equipment is designed for the professional use in ground stations and in cable TV head end stations.

DEV Systemtechnik GmbH declares that this equipment meets all relevant standards and rules. The unit carries a CE mark.

Please read all instructions before installation or usage of the equipment!

2.1 Warranty

The current warranty terms and conditions are stated on the company homepage, please refer to www.dev-systemtechnik.com.

2.2 Disclaimer

The information contained herein is believed to be reliable. DEV Systemtechnik makes no warranties regarding the information contained herein.
All information contained herein is subject to change without notice.

2.3 Warnings

The following safety measures must be observed during all phases of operation, service, and repair of the equipment. Failure to comply with the safety precautions or warnings in this document violates safety standards of design, manufacturing, and intended use of the equipment and may affect the built-in protections.
DEV Systemtechnik shall not be liable for the failure of users to comply with these requirements.
Disconnect the equipment from any external power line before opening of the housing. In any case, the equipment is to be opened by trained service staff, only.

Environmental Conditions
The equipment is designed for indoor use for an ambient temperature range of 0…+45 °C (32…113 °F).

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Mechanical Integration

It is not permitted to fix the 1 RU chassis on the rack mount flanges, only. Balanced weight distribution of the chassis has to be ensured by placing it on a solid surface or by using rack rails.

Sufficient airflow through the chassis must be ensured to achieve best cooling results. Do not block or obstruct the airflow through the chassis at the front and at the rear side e.g., by placing large components close to the rear side. It is recommended to provide 1 RU space in the rack above and below the chassis to ensure a good air circulation.

Heat Hazard

Top, bottom, and side surfaces may become hot when operating the equipment continuously. To reduce the risk of injury from a hot surface, allow the surface to cool before touching.

Grounding, AC Connection, Cables

To reduce the risk of shock hazard, the chassis must be connected to an electrical ground via a ground wire, please refer to chapter 4.2.2.

For the AC connection, each AC power connector of the chassis must be connected to the AC mains supply through a power cable with three conductors and with the ground wire firmly connected to an electrical ground (safety ground) at the power outlet. Any interruption of the protective ground conductor will cause a potential shock hazard that might cause personal injury.

Risk of electrical shock and energy hazard

Disconnecting a single power supply line only does not interrupt the AC supply of the equipment. To isolate the chassis completely, disconnect all power supply lines of all AC connectors.

Do not connect the equipment to mains supply exceeding the specifications for input voltage and frequency. Recommended AC cables to be applied are standard high temperature power cables with type IEC 60320-C13 appliance plug appropriately rated to the conditions of the country where the equipment is operated.

LNB Power

The DEV 2151 amplifiers and the DEV 214x/DEV 215x distribution amplifiers are capable to deliver LNB power in four different ways/modes:

"VL" (vertical low: 13 V, 0 Hz), "VH" (vertical high: 13 V, 22 kHz), "HL" (horizontal low: 18 V, 0 Hz), or "HH" (horizontal high: 18 V, 22 kHz).

All four LNB power supply modes are available via the (related) front side amplifier USB interface (chapter 5.1.10), via Web Interface (chapter 5.2.3.1), and via SNMP (chapter 5.3.4.3.3.8).

It is recommended to turn off the LNB power while working on the RF cabling to avoid damages. Otherwise, there is a risk that a high impulse current degrades the RF performance of the corresponding RF input connector.

It is not permitted to supply a current sink (e.g., an LNB) with the LNB power supply of more than one (distribution) amplifier channel (i.e., parallel connection) for several reasons:

- If turned on with different voltages i.e., 13 V (vertical) and 18 V (horizontal), the higher voltage feeds backward the LNB power supply of the (distribution) amplifier channel(s) providing the lower voltage and possibly destroys the LNB power supply and/or coils of a related splitter.

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- If turned on in a "high" LNB power supply mode i.e., with a 22 kHz tone signal modulated onto the DC voltage, the signal waves are superimposed and may even cancel out each other, since the 22 kHz generation of different (distribution) amplifier channels is not synchronized.
- Even if switched to the same LNB power supply mode without a 22 kHz tone (VL, HL), it is not recommended to supply a current sink with the LNB power supply of more than one (distribution) amplifier channel. Due to variations of parts, the current load for the (distribution) amplifier channels is not balanced and thus the current monitoring functionality is not working properly.

3 Equipment Description

With the DEV 2x4x/DEV 2x5x series, DEV Systemtechnik offers eighteen different managed/redundant distribution/combining amplifier products in a rack mountable 1 RU chassis:

- DEV 2142 1:8 Managed Distribution Amplifier; 800...2450 MHz
- DEV 2142/N 1:8 Managed Distribution Amplifier; 800...2450 MHz; 50 Ohm, N (f)
- DEV 2143 2*1:8 Managed Distribution Amplifier; 800...2450 MHz
- DEV 2145 1:16 Managed Distribution Amplifier; 800...2450 MHz

- DEV 2151 Redundant Amplifier; 800...2450 MHz
- DEV 2151/N Redundant Amplifier; 800...2450 MHz; 50 Ohm, N (f)

- DEV 2152 1:8 Redundant Distribution Amplifier; 800...2450 MHz
- DEV 2152/N 1:8 Redundant Distribution Amplifier; 800...2450 MHz; 50 Ohm, N (f)
- DEV 2153 2*1:8 Redundant Distribution Amplifier; 800...2450 MHz
- DEV 2155 1:16 Redundant Distribution Amplifier; 800...2450 MHz

- DEV 2242 8:1 Managed Combining Amplifier; 800...2450 MHz
- DEV 2242/N 8:1 Managed Combining Amplifier; 800...2450 MHz; 50 Ohm, N (f)
- DEV 2243 2*8:1 Managed Combining Amplifier; 800...2450 MHz
- DEV 2245 16:1 Managed Combining Amplifier; 800...2450 MHz

- DEV 2252 8:1 Redundant Combining Amplifier; 800...2450 MHz
- DEV 2252/N 8:1 Redundant Combining Amplifier; 800...2450 MHz; 50 Ohm, N (f)
- DEV 2253 2*8:1 Redundant Combining Amplifier; 800...2450 MHz
- DEV 2255 16:1 Redundant Combining Amplifier; 800...2450 MHz

Via product version options, highest configuration flexibility is provided by selecting the impedance (50 Ohm or 75 Ohm) and the RF connector type (SMA (f), BNC (f), or F (f)) for the input ports and the output ports when ordering a DEV 2x4x/DEV 2x5x (except the 50 Ohm N-connector products DEV 2xxx/N, see below):

As a default, the redundant amplifier DEV 2151 is equipped with a single amplifier, by ordering the DEV 2151 with Option 29, the product provides two amplifiers.

- <u>per amplifier input port</u> an amplifier input port option needs to be specified
  - Option vAin50 One Amplifier Input Port 50 Ohm, SMA (f)
  - Option vAin50B One Amplifier Input Port 50 Ohm, BNC (f)
  - Option vAin75 One Amplifier Input Port 75 Ohm, F (f)
  - Option vAin75B One Amplifier Input Port 75 Ohm, BNC (f)

and

- <u>per amplifier output port</u> an amplifier output port option needs to be specified
  - Option vAout50 One Amplifier Output Port 50 Ohm, SMA (f)
  - Option vAout50B One Amplifier Output Port 50 Ohm, BNC (f)
  - Option vAout75 One Amplifier Output Port 75 Ohm, F (f)
  - Option vAout75B One Amplifier Output Port 75 Ohm, BNC (f)

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For the distribution amplifiers DEV 214x/DEV 215x,

- per splitter input port a splitter input port option needs to be specified
  - Option vSin50 One Splitter Input Port 50 Ohm, SMA (f)
  - Option vSin50B One Splitter Input Port 50 Ohm, BNC (f)
  - Option vSin75 One Splitter Input Port 75 Ohm, F (f)
  - Option vSin75B One Splitter Input Port 75 Ohm, BNC (f)

and

- per 8 splitter output ports a splitter output port option needs to be specified
  - Option vSout50 Eight Splitter Output Ports 50 Ohm, SMA (f)
  - Option vSout50B Eight Splitter Output Ports 50 Ohm, BNC (f)
  - Option vSout75 Eight Splitter Output Ports 75 Ohm, F (f)
  - Option vSout75B Eight Splitter Output Ports 75 Ohm, BNC (f)

For the combining amplifiers DEV 224x/DEV 225x,

- per 8 combiner input ports a combiner input port option needs to be specified
  - Option vCin50 Eight Combiner Input Ports 50 Ohm, SMA (f)
  - Option vCin50B Eight Combiner Input Ports 50 Ohm, BNC (f)
  - Option vCin75 Eight Combiner Input Ports 75 Ohm, F (f)
  - Option vCin75B Eight Combiner Input Ports 75 Ohm, BNC (f)

and

- per combiner output port a combiner output port option needs to be specified
  - Option vCout50 One Combiner Output Port 50 Ohm, SMA (f)
  - Option vCout50B One Combiner Output Port 50 Ohm, BNC (f)
  - Option vCout75 One Combiner Output Port 75 Ohm, F (f)
  - Option vCout75B One Combiner Output Port 75 Ohm, BNC (f)

For the 50 Ohm N-connector products (DEV 2151/N, DEV 2142/N, DEV 2241/N, DEV 2152/N, DEV 2251/N) no product version options need to be specified, for the amplifier DEV 2151/N only Option 29 is offered to possibly order the product with two amplifiers installed.

All DEV 2x4x/DEV 2x5x provide redundant power supplies; and -for each channel- a monitoring output port, RF level monitoring functionality, and variable gain and variable slope functionality.
Additionally, the DEV 214x/DEV 215x (distribution) amplifiers feature an LNB power supply with current monitoring (for each amplifier channel).

Chapter 3.5 illustrates a few equipment configurations.

Next, the operating panel of the amplifiers is described. For the communication features, please refer to chapter 3.2. The redundant amplifier feature of the DEV 2x5x is outlined in chapter 3.3.

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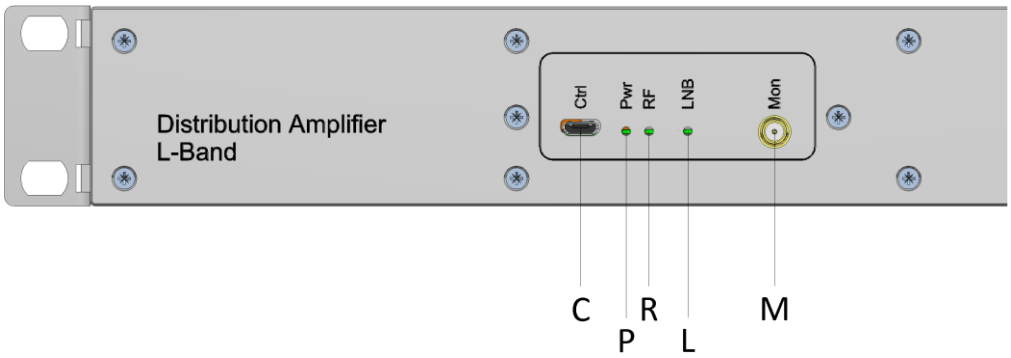
3.1 Amplifier Operating Panel

As indicated above, the DEV 2x4x/DEV 2x5x provides one or two amplifiers at the front side. The operating panel of each channel provides only a few elements:



Each front side amplifier features an USB interface labeled "**Ctrl**" (**C**) providing elementary control; please refer to chapter 5.1.

The power LED labeled "**Pwr**" (**P**) is usually on and green (🟢) if the front side amplifier is supplied with power and if the amplifier circuits are working normally. The LED turns to red (🔴) if amplifier circuits are detected as being defective. Additionally, the power LED -in case of a product with two channels: only the power LED of the left front side amplifier- is used for power line monitoring:

- If the first power line (i.e., the left primary connector on the rear side) is not connected, the LED blinks in red.
- If the second power line (i.e., the right primary connector on the rear side) is not connected, the LED blinks twice in red.
- If the chassis is supplied via both power lines, the behavior of the LED is as described above.

With the RF level monitoring functionality of each channel, the aggregated signal power level of the RF signal is measured and compared with an RF threshold level. The status is signalized via the "**RF**" LED (**R**): Green (🟢) means that the signal level is above the threshold level and red (🔴) signalizes an alarm (i.e., the signal level is below the threshold level).

Only for the DEV 214x/DEV 215x (distribution) amplifiers, the LED labeled "**LNB**" (**L**) is available signalizing the status of the LNB power supply: If the LNB power supply is turned off, the LED is off. If the LNB power supply is turned on and if a current is fed to the input of the (distribution) amplifier that is within a certain adjustable range (i.e., the current monitoring interval), the LED is green (🟢). If the current delivered by the LNB power supply is out of range, the LED is red (🔴), signalizing an alarm.

Finally, the SMA port labeled "**Mon**" (**M**) of each front side amplifier can be used for the input signal monitoring of an amplifier, of a distribution amplifier, or for the combined input signal monitoring of a combining amplifier at a reduced signal level.

3.2 Communication Features

In addition to the above mentioned front side amplifier USB interface, there are implemented further communication features for the monitoring and for the control of the DEV 2x4x/DEV 2x5x.

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The Web Interface (chapter 5.2) is intended to be the main local user interface for operating the DEV 2x4x/DEV 2x5x; it provides full control in terms of checking the health status and of setting up specific parameters. The Web Interface is accessible through a web browser via the Ethernet interface.

The implementation of SNMP (Simple Network Management Protocol) via the Ethernet interface permits remote monitoring and control of the equipment (chapter 5.3).

3.3 Redundant Amplifiers

The DEV 2151 amplifiers, the DEV 215x distribution amplifiers, and the DEV 225x combining amplifiers are equipped with one or two redundant amplifiers. I.e., each amplifier provides a redundant amplifier path that can be operated in cold or hot standby.
In case that a failure of the main amplifier path is detected by the applied amplifier current monitoring functionality, the switching from the main amplifier path to the redundant amplifier path is executed automatically.

3.4 Other Options

By default, eight Web Interface licenses are included in the delivery; additional licenses for the Web Interface can be ordered at any time, please contact DEV Systemtechnik.
With applied Option 77, the Web Interface of the DEV 2x4x/DEV 2x5x provides the Recording Window (please refer to chapter 5.2.4) that permits the visualization and the external storage of amplifier signal data.

3.5 Equipment Configurations

As stated above, the product version options permit highest flexibility by specifying the impedance (50 Ohm or 75 Ohm) and the RF connector type (SMA (f), BNC (f), or F (f)) for the input ports and the output ports of a DEV 2x4x/DEV 2x5x.
The appearance of the DEV 2x4x/DEV 2x5x is similar, there are to be distinguished four different front sides, for the different rear sides, eight examples are shown in order to demonstrate the versatility how the chassis can be set up.

3.5.1 Front Side Single (Distribution) Amplifier

This front side is installed for the DEV 2151(/N) (equipped with a single amplifier), for the DEV 2142(/N), DEV 2145, DEV 2152(/N), and for the DEV 2155.

Front Side

C	"Ctrl" control connector, Micro USB
P	"Pwr" power LED
R	"RF" LED
L	"LNB" power LED
M	"Mon" monitor port, 50 Ohm, SMA (f)

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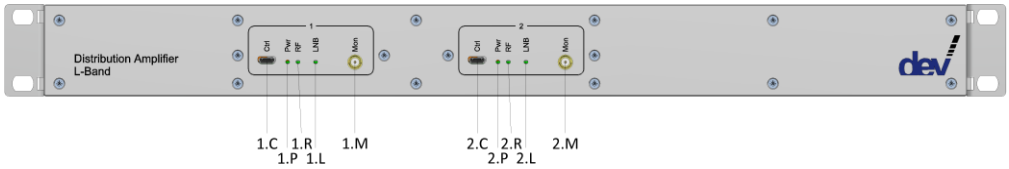
3.5.2 Front Side Dual (Distribution) Amplifier

This front side is installed for the DEV 2151(/N) (equipped with two amplifiers), for the DEV 2143, and for the DEV 2153.

Front Side

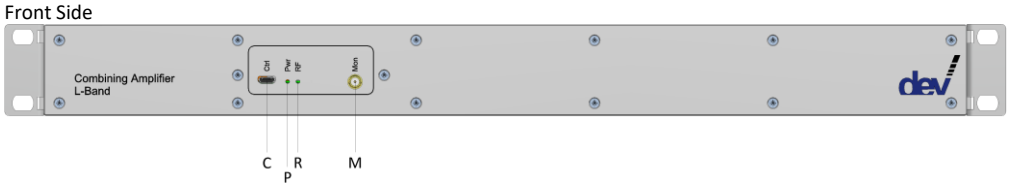


1.C, 2.C	"Ctrl" control connector, Micro USB
1.P, 2.P	"Pwr" power LED
1.R, 2.R	"RF" LED
1.L, 2.L	"LNB" power LED
1.M, 2.M	"Mon" monitor port, 50 Ohm, SMA (f)

3.5.3 Front Side Single Combining Amplifier

This front side is installed for the DEV 2242(/N), DEV 2245, DEV 2252(/N), and for the DEV 2255.

Front Side

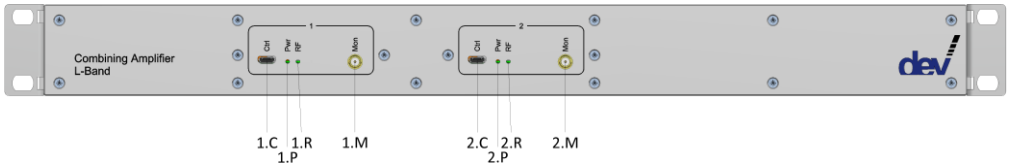


C	"Ctrl" control connector, Micro USB
P	"Pwr" power LED
R	"RF" LED
M	"Mon" monitor port, 50 Ohm, SMA (f)

3.5.4 Front Side Dual Combining Amplifier

This front side is installed for the DEV 2243 and for the DEV 2253.

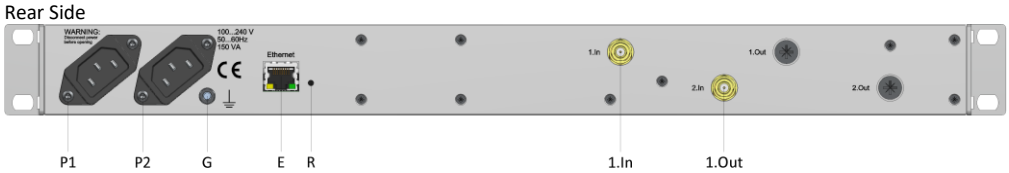
Front Side



1.C, 2.C	"Ctrl" control connector, Micro USB
1.P, 2.P	"Pwr" power LED
1.R, 2.R	"RF" LED
1.M, 2.M	"Mon" monitor port, 50 Ohm, SMA (f)

3.5.5 Rear Side DEV 2151 (single amplifier)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

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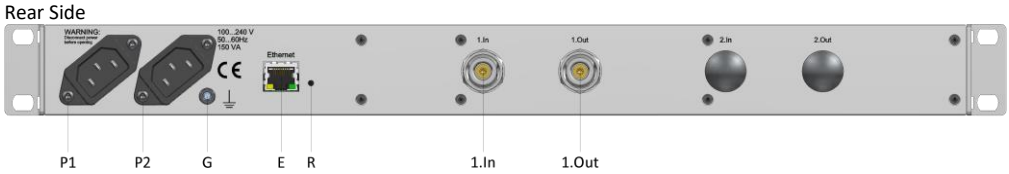
Amplifier Configuration

1 * DEV 2151
1 * Option vAin50
1 * Option vAout50

1.In	"1.In" RF input port equipped with 50 Ohm, SMA (f)
1.Out	"1.Out" RF output port equipped with 50 Ohm, SMA (f)

3.5.6 Rear Side DEV 2151/N (single amplifier)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

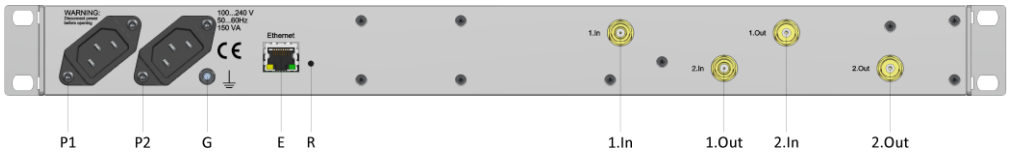
Amplifier Configuration

1 * DEV 2151/N

1.In	"1.In" RF input port equipped with 50 Ohm, N (f)
1.Out	"1.Out" RF output port equipped with 50 Ohm, N (f)

3.5.7 Rear Side DEV 2151 (dual amplifier)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

Amplifier Configuration

1 * DEV 2151
1 * Option vAin50
1 * Option vAout50
1 * Option 29
1 * Option vAin75
1 * Option vAout75

1.In	"1.In" RF input port equipped with 50 Ohm, SMA (f)
1.Out	"1.Out" RF output port equipped with 50 Ohm, SMA (f)
2.In	"2.In" RF input port equipped with 75 Ohm, F (f)
2.Out	"2.Out" RF output port equipped with 75 Ohm, F (f)

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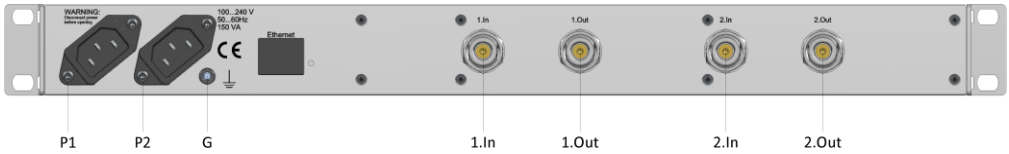
3.5.8 Rear Side DEV 2151/N (dual amplifier)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

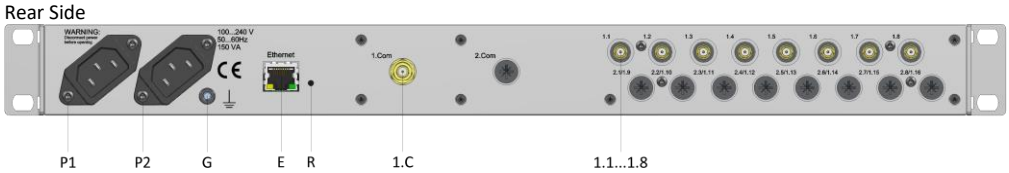
Amplifier Configuration

1 * DEV 2151/N
1 * Option 29

1.In	"1.In" RF input port equipped with 50 Ohm, N (f)
1.Out	"1.Out" RF output port equipped with 50 Ohm, N (f)
2.In	"2.In" RF input port equipped with 50 Ohm, N (f)
2.Out	"2.Out" RF output port equipped with 50 Ohm, N (f)

3.5.9 Rear Side DEV 2142 (DEV 2152) or DEV 2242 (DEV 2252)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

Distribution Amplifier Configuration

1 * DEV 2142 (or DEV 2152)
1 * Option vSin50
1 * Option vSout50

1.C	"1.Com" RF input port equipped with 50 Ohm, SMA (f)
1.1...1.8	"1.1"..."1.8" RF output ports equipped with 50 Ohm, SMA (f)

Combining Amplifier Configuration

1 * DEV 2242 (or DEV 2252)
1 * Option vCin50
1 * Option vCout50

1.C	"1.Com" RF output port equipped with 50 Ohm, SMA (f)
1.1...1.8	"1.1"..."1.8" RF input ports equipped with 50 Ohm, SMA (f)

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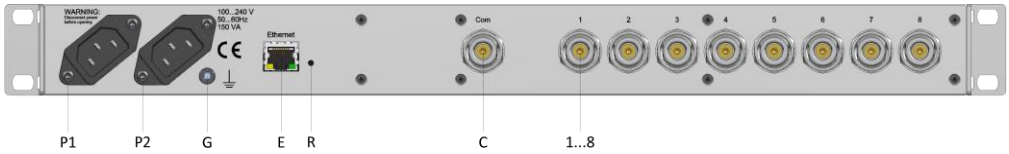
3.5.10 Rear Side DEV 2142/N (DEV 2152/N) or DEV 2242/N (DEV 2252/N)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

Distribution Amplifier Configuration

1 * DEV 2142/N (or DEV 2152/N)

C	"Com" RF input port equipped with 50 Ohm, N (f)
1…8	"1"…"8" RF output ports equipped with 50 Ohm, N (f)

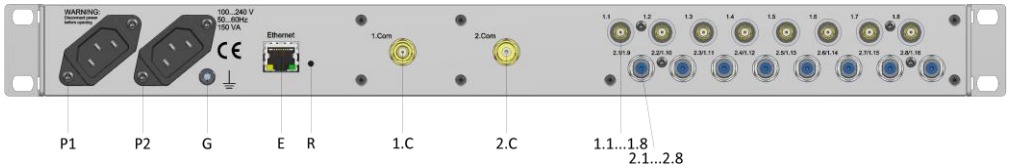
Combining Amplifier Configuration

1 * DEV 2242/N (or DEV 2252/N)

C	"Com" RF output port equipped with 50 Ohm, N (f)
1…8	"1"…"8" RF input ports equipped with 50 Ohm, N (f)

3.5.11 Rear Side DEV 2143 (DEV 2153) or DEV 2243 (DEV 2253)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

Distribution Amplifier Configuration

1 * DEV 2143 (or DEV 2153)
1 * Option vSin50
1 * Option vSout50
1 * Option vSin75
1 * Option vSout75

1.C	"1.Com" RF input port equipped with 50 Ohm, SMA (f)
2.C	"2.Com" RF input port equipped with 75 Ohm, F (f)
1.1…1.8	"1.1"…"1.8" RF output ports equipped with 50 Ohm, SMA (f)
2.1…2.8	"2.1"…"2.8" RF output ports equipped with 75 Ohm, F (f)

Combining Amplifier Configuration

1 * DEV 2243 (or DEV 2253)
1 * Option vCin50
1 * Option vCout50
1 * Option vCin75
1 * Option vCout75

1.C	"1.Com" RF output port equipped with 50 Ohm, SMA (f)
2.C	"2.Com" RF output port equipped with 75 Ohm, F (f)
1.1…1.8	"1.1"…"1.8" RF input ports equipped with 50 Ohm, SMA (f)
2.1…2.8	"2.1"…"2.8" RF input ports equipped with 75 Ohm, F (f)

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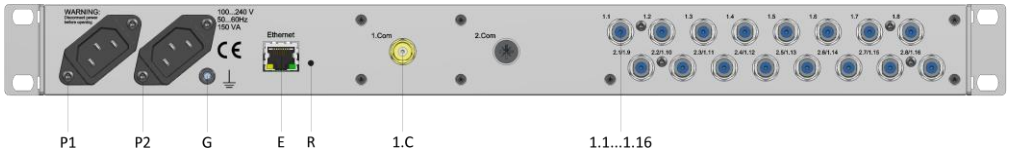
3.5.12 Rear Side DEV 2145 (DEV 2155) or DEV 2245 (DEV 2255)

Rear Side



P1	AC plug for power supply unit 1
P2	AC plug for power supply unit 2
E	"Ethernet" connector, RJ-45
R	Reset button
G	Grounding bolt

Distribution Amplifier Configuration

1 * DEV 2145 (or DEV 2155)
1 * Option vSin75
2 * Option vSout75

1.Com	"1.Com" RF input port equipped with 75 Ohm, F (f)
1.1…1.16	"1.1"…"1.16" RF output ports equipped with 75 Ohm, F (f)

Combining Amplifier Configuration

1 * DEV 2245 (or DEV 2255)
1 * Option vCin75
2 * Option vCin75

4	"1.Com" RF output port equipped with 75 Ohm, F (f)
1.1…1.16	"1.1"…"1.16" RF input ports equipped with 75 Ohm, F (f)

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4 Installation Instructions

4.1 Scope of Delivery

1 * DEV 2142, DEV 2142/N, DEV 2143, DEV 2145, DEV 2151, DEV 2151/N, DEV 2152, DEV 2152/N, DEV 2153, DEV 2155, DEV 2242, DEV 2242/N, DEV 2243, DEV 2245, DEV 2252, DEV 2252/N, DEV 2253, or DEV 2255 *equipped with (product version) options as ordered*

1 * User Manual (this document)

4.2 Installation of the Equipment

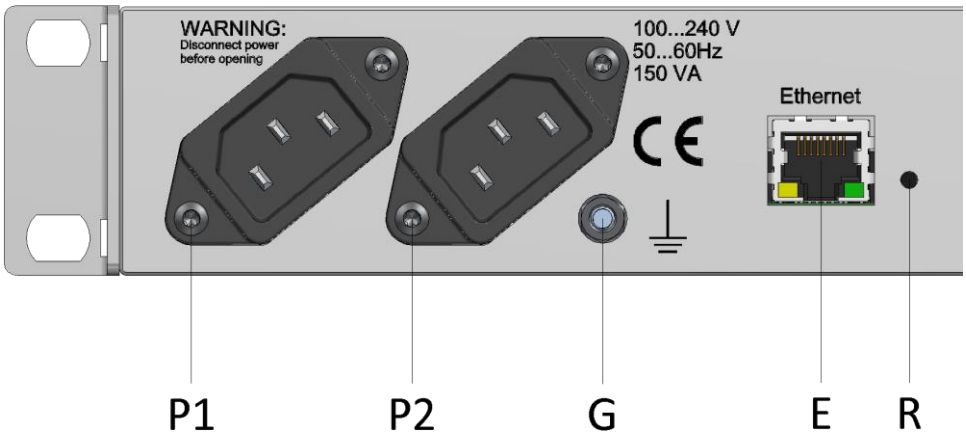
4.2.1 Mechanical Integration

Please consider the "Mechanical Integration" warnings in chapter 2.3!
For the integration in a 19" rack, the rack slots must be prepared with rails for the chassis. After inserting the chassis in the rack, fix the chassis with four screws to the rack at its rack mount flanges.

Note:

- The rack mounting ears can be unscrewed, thus the chassis can be shifted back and forth on the rack rails in order to facilitate the assembly of cables.

4.2.2 Grounding, Data Cable, and Power Connection



Please consider the "Grounding, AC Connection, Cables" warnings in chapter 2.3!

The chassis needs to be connected to the 19" rack via a ground wire. The grounding bolt (labeled **G** in the figure) is located on the left at the rear side of the chassis. Take off the upper nut and the first washer of the grounding bolt and connect the grounding cable, which must have a ring tongue terminal matching the M4 fastening bolt. After that, put on the washer and the nut again and tighten the nut.

Establish the external Ethernet connection by plugging an appropriate Ethernet cable from the target network (chapter 4.4.2.1) to the Ethernet connector (**E**) that is located on the left at the rear side of the chassis.

Finally, connect appropriate power cords to the primary connectors (**P1** & **P2**). It is suggested to use two separately fused phases of the electrical outlets.

Note:

- It is **highly recommended** to power up the equipment after the installation of all signal cables (chapter 4.2.3)

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4.2.3 Connection of RF Signal Cables

For the DEV 214x/DEV 215x (distribution) amplifiers, please consider the "LNB power" warnings in chapter 2.3!

Apply strain relieves to the external RF signal cables which are to be connected in a way, that the connectors of the chassis are not stressed mechanically.

When connecting RF cables to the chassis, each connector providing a thread is to be tightened with a torque wrench. The torque applied to 75 Ohm F connectors should not exceed 4.5 Nm (39.8 lbf-in). In case of 50 Ohm SMA connectors a torque wrench adjusted to 1 Nm (8.85 lbf-in) is to be used. If equipped with BNC connectors, make sure that the bayonet of each BNC connector is locked.

Note:

- If ordered with product version options (chapter 3) providing (75 Ohm female) F connectors, make sure, that the applied connectors are male precision F connectors (with thin inner conductor).
- If ordered with product version options providing (50 Ohm or 75 Ohm female) BNC connectors, make sure, that the applied connectors are male BNC connectors with a matching impedance.
- For best RF performance, connect all ports appropriately and terminate unused ports with an appropriate load.
- For distribution amplifiers best signal quality is achieved by connecting the signal receivers symmetrically to the output ports
  - 1…4 & 5…8 of each 1:8 splitter (DEV 2142, DEV 2142/N, DEV 2143, DEV 2152, DEV 2152/N, DEV 2153)
  - 1…4, 5…8, 9…12, and 13…16 of the 1:16 splitter (DEV 2145, DEV 2155)

  and by connecting impedance matched loads to the unused output ports
- For combining amplifiers best signal quality is achieved by connecting the signal receivers symmetrically to the input ports
  - 1…4 & 5…8 of each 8:1 combiner (DEV 2242, DEV 2242/N, DEV 2243, DEV 2252, DEV 2252/N, DEV 2253)
  - 1…4, 5…8, 9…12, and 13…16 of the 16:1 combiner (DEV 2245, DEV 2255)

  and by connecting impedance matched loads to the unused input ports
- If all cables are connected, the equipment can be powered up. Now (for (distribution) amplifiers), the LNB power can be turned on for the front side amplifier(s) that is (are) required to feed power e.g., to an LNB.
  Again, please consider the "LNB power" warnings in chapter 2.3!

4.3 Equipment Factory Settings

The equipment is delivered with factory settings as described in the following.
It is possible to restore these settings, please refer to chapters 5.1.12 & 5.2.6.3.1.

4.3.1 Factory Settings for the Front Side Amplifiers

Each front side amplifier provides RF level monitoring functionality i.e., the aggregated power level of the RF signal is measured and compared with an RF threshold level. If the signal level is below the threshold level, an event is generated. If the signal level is below the threshold level, an alarm is triggered which is indicated via LED, via Web Interface (please refer to chapter 5.2.3), or via SNMP (chapter 5.3.4.3.3.18).

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The factory setting for the RF threshold level is:

RF Threshold Level	*-30 dBm*

The gain setting of each front side amplifier can be adjusted. The factory setting is:

Gain	*0 dB*

The slope setting of each front side amplifier can be adjusted. The factory setting is:

Slope	*0 dB*

All DEV 214x/DEV 215x (distribution) amplifiers are capable to provide LNB power at the related RF input port. With the current monitoring functionality, the current delivered by the RF input port is monitored. If the measured current exceeds the defined monitoring interval, an event is generated. The factory settings for the LNB power mode and for the current monitoring interval are:

LNB Power Mode	*Off*
Lower Limit	*100 mA*
Upper Limit	*250 mA*

Note:

- All settings can be adjusted via the (related) USB interface, via Web Interface, and via SNMP

4.3.2 Factory Network Settings

The factory network settings are:

IP-Address	*192.168.0.11*
Netmask	*255.255.255.0*
Gateway	*192.168.0.99*

For changing the network setup, please refer to chapter 5.2.6.2.1.

4.3.3 Factory SNMP Settings

The factory SNMP settings are configured as follows:

Read Community	*public*
Write Community	*private*

Additionally, the default community for all four *Trap Destination IP Addresses* is set to *public* and all configurable traps are *disabled*. If an address is defined without explicit port assignment, the standard SNMP trap port *162* is assigned implicitly.

For changing the SNMP setup, please refer to chapter 5.2.6.2.3.

There are software factory defaults for the following SNMP objects:

- .application.portTable.portEntry.poEnable chapter 5.3.4.3.3.2
- .application.portTable.portEntry.poRfLimit chapter 5.3.4.3.3.6
- .application.portTable.portEntry.poBiasOnOff chapter 5.3.4.3.3.8
- .application.portTable.portEntry.poBiasLimitLo chapter 5.3.4.3.3.10
- .application.portTable.portEntry.poBiasLimitHi chapter 5.3.4.3.3.11
- .application.portTable.portEntry.poGain chapter 5.3.4.3.3.13
- .application.portTable.portEntry.poTilt chapter 5.3.4.3.3.15
- .surveillance.trapEnableMask chapter 5.3.4.4.1
- .surveillance.errorEnableMask chapter 5.3.4.4.2

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4.4 Communication Interface Configuration

Two types of hardware interfaces are provided: one (or two) USB interface(s) and an Ethernet interface.

4.4.1 Front Side Amplifier USB Interface

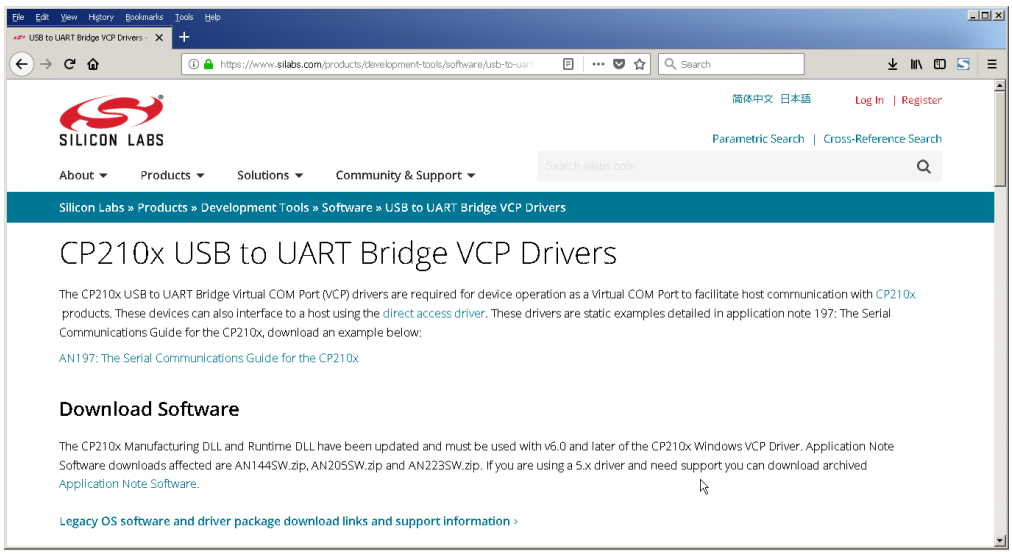
Each channel features a Micro USB receptacle to provide elementary control

- for the RF level monitoring functionality,
- for the variable gain functionality, and
- for the variable slope functionality
- For the DEV 214x/DEV 215x (distribution) amplifiers, additionally the operation mode of the LNB power supply and the setup of the LNB current monitoring alarm interval can be controlled (per channel)

Finally, it is possible to restore the factory default settings (chapter 4.3). Chapter 5.1 describes the operation via the front side amplifier USB interface.

4.4.1.1 USB Driver Installation and Configuration

The applied USB interface is a CP210x USB to UART Bridge Virtual COM Port by Silicon Laboratories, Inc. This requires a driver to be downloaded, to be installed, and to be configured on your computer. Currently, the driver download page is https://www.silabs.com/products/development-tools/software/usb-to-uart-bridge-vcp-drivers



After selecting and downloading the appropriate driver archive, unpack and install the driver.

Next, connect the computer to a/the front side amplifier USB interface using a fully populated USB cable with an USB-A connector (2.0 or higher) on one end and a Micro-USB-B connector on the other end.

When using a MS Windows operating system (this is assumed in the following), the detection and installation of the hardware is performed automatically.

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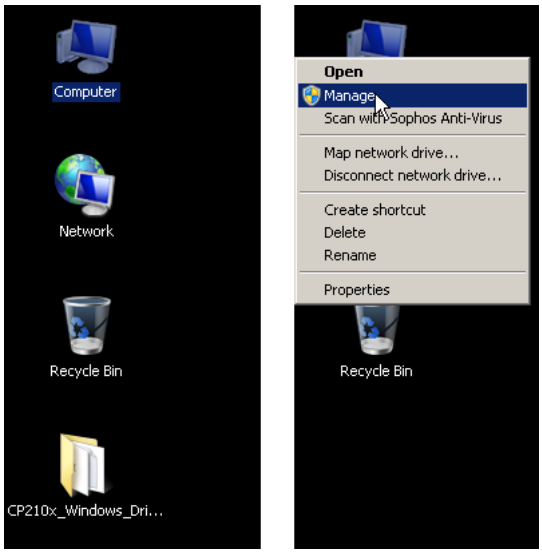
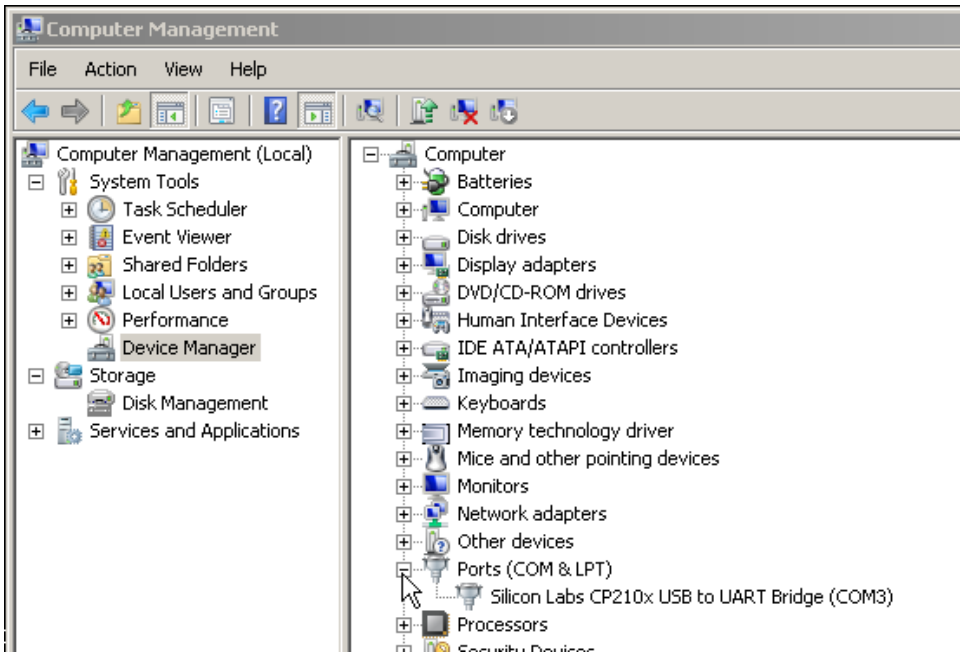
To identify (or to configure) the COM port number and to (optionally) configure the port settings ("Bits per second", "Data bits", "Parity", "Stop bits", and "Flow control") of the amplifier USB interface, locate the "Computer" icon on your desktop (or within the Windows Explorer), open the context menu (right click), and select "Manage":



After the Computer Management window has appeared, select the "Device Manager" on the left side and expand "Ports (COM & LPT)" on the right side:



The available COM ports appear (here just the front side amplifier USB interface) with the COM port number in parentheses: "Silicon Labs CP210x USB to UART Bridge (COM3)" (note that the COM port number may be different, here COM3 is assigned)
Except the COM port number assignment, the port settings are adjusted implicitly by configuring the terminal application (chapter 4.4.1.2) appropriately i.e., it is not required to configure the port settings explicitly within the "Device Manager" of the Computer Management window

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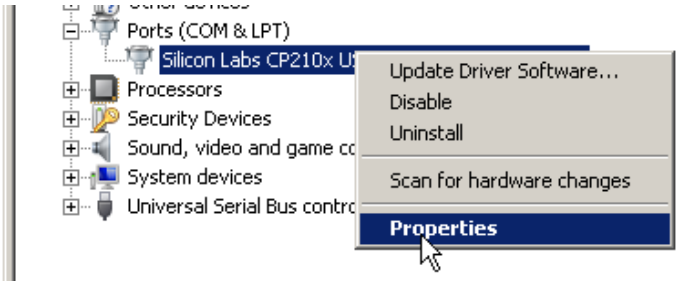
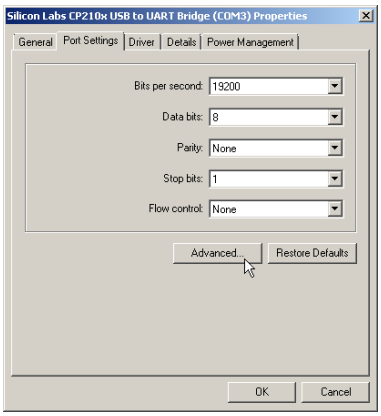
The port settings and the COM port number assignment of the interface can be altered by selecting (here) the "Silicon Labs CP210x USB to UART Bridge (COM3)" and by selecting "Properties" within the context menu (right click)



Select the "Port Settings" tab and (optionally) apply the settings as shown below. Note that the COM port number assignment can be altered when clicking on Advanced...



Note:

- When connecting a front side amplifier USB interface to a different computer USB port, there is no need to install the driver again.
  However, since MS Windows assigns a new COM port number if a different computer USB port (with an already detected front side amplifier USB interface) is used, it is necessary at least to check the COM port number assignment of the interface. It is possible to alter the COM port number assignment for each computer USB port (to the same COM port) via the Computer Management window as described above.

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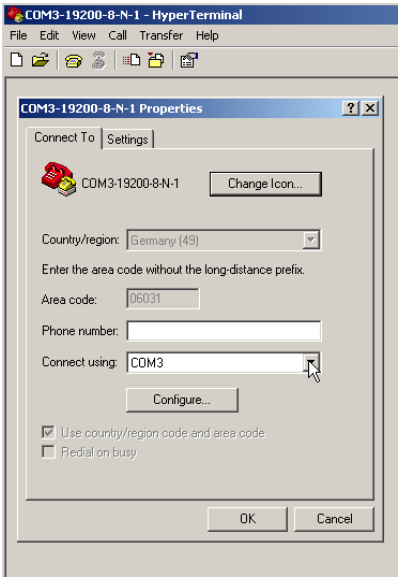
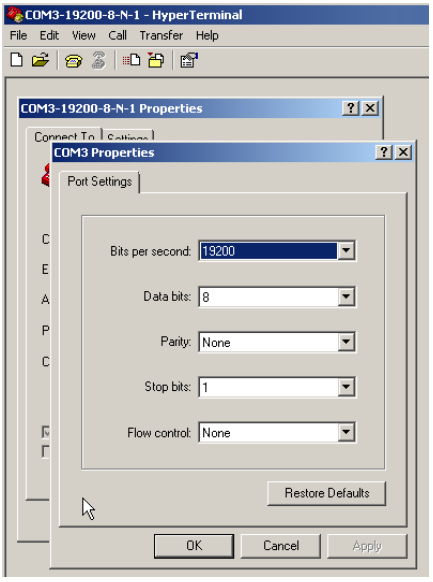
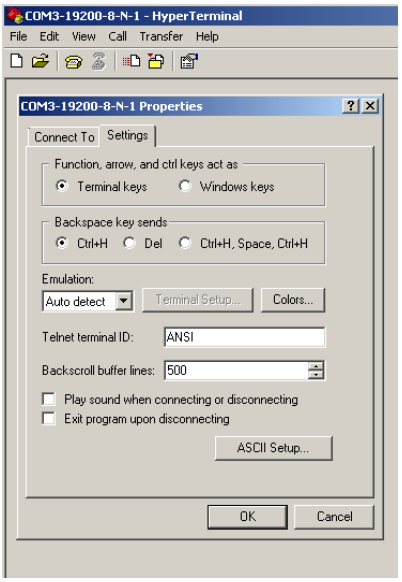
4.4.1.2 Terminal Application

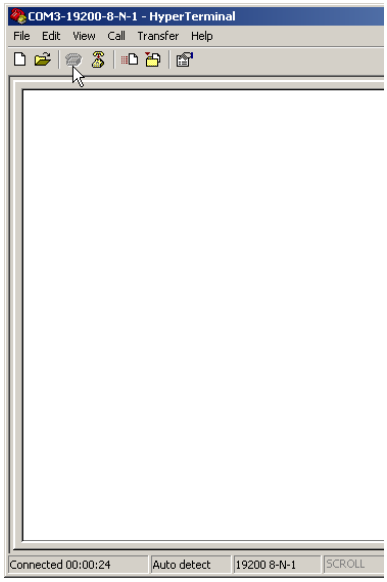
For the operation via the amplifier USB interface, a terminal application is required e.g., MS Windows HyperTerminal. The parameters of the terminal application need to be configured appropriately.
If using MS Windows HyperTerminal and if being disconnected -by clicking on the right phone icon ( ) -, choose the "Properties" entry from the "File" menu and select the appropriate COM port (here COM3, please consider the note in chapter 4.4.1.1) in the drop down list labeled "Connect using:" (left figure). After clicking on the Configure... button, adjust the port settings as stated in the right figure ("Bits per second": 19200, "Data bits": 8, "Parity": None, "Stop bits": 1, and "Flow control": None)





After that, click on the "Settings" tab. There, select "Auto detect" from the "Emulation" drop down list and then click on the ASCII Setup... button to uncheck "Send line ends with line feeds" (left figure). Finally connect the HyperTerminal by clicking on the left phone icon ( ), the connection status is shown in the status line at the bottom of the window (right figure)





4.4.2 Ethernet Interface

For the external communication, an RJ-45 jack is provided at the rear side of the chassis. This standard 100Base-TX Ethernet interface is used for the operation via Web Interface or via SNMP.

4.4.2.1 Ethernet Connection

If a standalone DEV 2x4x/DEV 2x5x is to be connected to a computer only, the connection can be established via a direct Ethernet cable (or through a local hub/switch) between the Ethernet connector of the DEV 2x4x/DEV 2x5x and the Ethernet connector of the computer. The network setup of the computer is to be altered that it is in the same subnet as the DEV 2x4x/DEV 2x5x. If it is intended to use the factory network settings, the IP address setting of the standalone computer is to be set to 192.168.0.x, with x in the range 1…254, except 11, which is the address of the DEV 2x4x/DEV 2x5x e.g., set the IP address of the standalone computer to the default gateway address 192.168.0.99.
If the DEV 2x4x/DEV 2x5x is to be connected to the local network, make sure to change the network setup as required. Ask your network administrator for the details on network parameters which are to be used.
The factory network settings are subject of chapter 4.3.2.
The network setup can be changed via Web Interface (chapter 5.2.6.2.1).

4.5 Maintenance of the Equipment

The equipment is maintenance-free.
Cleaning of the chassis surface should be done carefully with a dry lint-free cloth.

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5 Operation of the Equipment

After going through the installation steps as described above, the equipment is ready to use. There are no elements to be operated at the chassis. The operating status is indicated via LEDs, which were already described in chapter 3.1.
Monitoring and control of the equipment can be performed via

- the Amplifier USB Interface (please refer to chapter 5.1)
- Web Interface (chapter 5.2) and/or
- SNMP (chapter 5.3)

5.1 Operation via the Front Side Amplifier USB Interface

As stated in chapter 4.4.1, each channel features a front side amplifier USB interface to provide elementary control. It is assumed here, that

- the USB driver is installed and configured (chapter 4.4.1.1), that
- the amplifier USB interface is connected to the computer, and that
- the terminal application is started and connected (chapter 4.4.1.2)

The single commands are explained in the following chapters. Some remarks:

- Except **`factory`** (chapter 5.1.12), all commands are issued by typing a single character (case insensitive)
- After the execution of all commands, a prompt character (">") signalizes that the next command can be issued.
- After issuing an inc/dec (increment/decrement) command (please refer to chapters 5.1.5…5.1.9), the current value of the selected parameter can be altered in defined increments using the "+" (plus) or "-" (minus) key.
  The inc/dec mode is finalized by typing the exclamation mark ("!") key.
  The inc/dec mode is activated again after issuing the next inc/dec command.
  Note that within inc/dec mode it is possible to issue other commands, typing "+" or "-" changes the parameter of the latest inc/dec command.
- Changes on all parameters that are performed via inc/dec commands are directly reflected in the (related) Amplifier Channel Window (chapter 5.2.3.1) of the Web Interface and in the SNMP .portTable (chapter 5.3.4.3.3)

5.1.1 Show Help Command (?)

Typing the "?" (question mark) key lists the available commands:

```
>?
?: show this help
0: show info
p: show current values
e: show error codes
r: inc/dec rf level threshold
g: inc/dec gain
s: inc/dec slope
u: inc/dec upper lnb-pwr threshold
l: inc/dec lower lnb-pwr threshold
   +/-: inc/dec selected value
   !: leave inc/dec mode
m: change lnb-pwr mode (off, VL, HL, VH, HH)
a: toggle redundant amp power (on/off)
factory: restore factory defaults
>
```

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Note:

- The commands i.e., lines starting with "`u`" (chapter 5.1.8), "`l`" (chapter 5.1.9), & "`m`" (chapter 5.1.10) appear only for DEV 214x/DEV 215x distribution amplifiers
- The command (i.e., line starting with) "`a`" (chapter 5.1.11) appears only for DEV 215x/DEV 225x redundant amplifiers.

5.1.2 Show Info Command (0)

Typing the "`0`" (zero) key lists information regarding hardware, software and capabilities of the amplifier

```
>0
hw: *13-0394* ** 42-0387 L-Band Red AmplifierS  version: 0 variant: 0
fw: 35-0145 A.06  build:  80 27 62 1C ** November 28 2019 09:00:27 R
capabilities:
- gain
- slope
- lnb-pwr
>
```

Note:

- Hardware/firmware information varies according to the applied amplifier type and firmware revision.
- The "`- lnb-pwr`" capability appears only for DEV 214x/DEV 215x (distribution) amplifiers.

5.1.3 Show Current Values Command (p)

Typing the "**`p`**" or "**`P`**" key lists measured values and current parameter settings:

```
>p
rf level (dBm | loThr): -50.0 | -30.0
amp path: main
red amp path power: off
gain (dB): 0.0
slope (dB): 1.0
lnb power (mode, mA, loThr, upThr): off0.0 100.0 250.0
>
```

Note:

- The line starting with "`rf level`" shows the currently measured RF level in "`dBm`" (here "`-50.0`") and the RF threshold level setting "`loThr`" (here "`-30.0`")
- The line starting with "`amp path`" and the line starting with "`red amp path power`" appear only for DEV 215x/DEV 225x redundant amplifiers. The line "`amp path`" states whether the redundant amplifier operates via the main or via the redundant amplifier path. The line "`red amp path power`" indicates whether the power supply of the redundant amplifier path is "`on`" or "`off`"
- The line starting with "`gain`" and the line starting with "`slope`" are indicating the current settings in "`dB`" of the variable gain (here "`2.0`") and slope (here "`1.0`") functionality.
- The line starting with "`lnb power`" appears only for DEV 214x/DEV 215x (distribution) amplifiers. The line states the operation mode ("`mode`") of the LNB power supply (here "`off`"), the currently measured LNB current in "`mA`" (here "`0.0`"), the lower "`loThr`" (here "`100.0`"), and the upper "`upThr`" (here "`250.0`") limit of the LNB current monitoring alarm interval.

5.1.4 Show Error Codes Command (e)

Typing the "**`e`**" or "**`E`**" key lists currently pending event(s)

```
>e
active errors: none
>
```


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5.1.5 Inc/Dec RF Level Threshold Command (r)

Typing the "r" or "R" key starts the inc/dec command to change the RF threshold level for the RF level monitoring functionality. The current value -as stated behind "curr. val:"- can be altered using the "+" (plus) or "-" (minus) key in increments of 0.5 dB. The inc/dec mode is finalized by typing the exclamation mark ("!") key.

```
>r
rf level threshold (curr. val:-30.0 ) use +/- to change
>+< -29.5
>+< -29.0
>+< -28.5
>-< -29.0
>-< -29.5
>-< -30.0
>!< leaving inc/dec mode
>
```

5.1.6 Inc/Dec Gain Command (g)

Typing the "g" or "G" key starts the inc/dec command to change the level of the variable gain functionality. The current value -as stated behind "curr. val:"- can be altered using the "+" (plus) or "-" (minus) key in increments of 0.5 dB. The inc/dec mode is finalized by typing the exclamation mark ("!") key.

```
>g
gain (curr. val:0.0 ) use +/- to change
>+< 0.5
>+< 1.0
>-< 0.5
>-< 0.0
>!< leaving inc/dec mode
>
```

5.1.7 Inc/Dec Slope Command (s)

Typing the "s" or "S" key starts the inc/dec command to change the level of the variable slope functionality. The current value -as stated behind "curr. val:"- can be altered using the "+" (plus) or "-" (minus) key in increments of 0.5 dB. The inc/dec mode is finalized by typing the exclamation mark ("!") key.

```
>s
slope (curr. val:1.0 ) use +/- to change
>+< 1.5
>+< 2.0
>-< 1.5
>-< 1.0
>!< leaving inc/dec mode
>
```

5.1.8 Inc/Dec Upper LNB Power Threshold Command (u)

This command is available for DEV 214x/DEV 215x (distribution) amplifiers, only. Typing the "u" or "U" key starts the inc/dec command to change the upper limit of the LNB current monitoring alarm interval. The current value -as stated behind "curr. val:"- can be altered using the "+" (plus) or "-" (minus) key in increments of 0.5 mA. The inc/dec mode is finalized by typing the exclamation mark ("!") key.

```
>u
upper lnb limit current (curr. val:250.0 ) use +/- to change
>+< 250.5
>+< 251.0
>+< 251.5
>-< 251.0
>-< 250.5
>-< 250.0
>!< leaving inc/dec mode
>
```

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5.1.9 Inc/Dec Lower LNB Power Threshold Command (l)

This command is available for DEV 214x/DEV 215x (distribution) amplifiers, only. Typing the "**l**" or "**L**" key starts the inc/dec command to change the lower limit of the LNB current monitoring alarm interval. The current value -as stated behind "curr. val:"- can be altered using the "+" (plus) or "-" (minus) key in increments of 0.5 mA. The inc/dec mode is finalized by typing the exclamation mark ("!") key.

```
>l
lower lnb limit current (curr. val:100.0 ) use +/- to change
>+< 100.5
>+< 101.0
>-< 100.5
>-< 100.0
>!< leaving inc/dec mode
>
```

5.1.10 Change LNB Power Mode Command (m)

This command is available for DEV 214x/DEV 215x (distribution) amplifiers, only. Each typing of "**m**" or "**M**" changes the operation mode of the LNB power supply. Thus, the available operation modes "VL", "VH", "HL", "HH", and "off" can be selected.

```
>m lnb power mode VL
>m lnb power mode VH
>m lnb power mode HL
>m lnb power mode HH
>m lnb power mode off
>
```

5.1.11 Toggle Redundant Amplifier Power (a)

This command is available for DEV 215x/DEV 225x redundant amplifiers, only. Each typing of "**a**" or "**A**" toggles the power supply of the redundant amplifier path (cold standby: off or hot standby: on)

```
>a
Redundant amp power is on
>a
Redundant amp power is off
>
```

5.1.12 Restore Factory Defaults Command (factory)

Typing the single letters of the string "**factory**" restores the equipment factory settings **completely** as described in chapter 4.3

```
>f
>a
>c
>t
>o
>r
>y
restore factory defaults
>
```

Note:

- After the confirmation "restore factory defaults" it takes ~60 seconds that the equipment is accessible again via Web Interface and via SNMP
- If the keys are not typed in the correct order, the command is aborted e.g.,

  ```
  >f
  >c
  unknown command
  >
  ```

- If it is desired to reset just the amplifier parameters, it is recommended to rather use the Factory Reset Window of the Web Interface (chapter 5.2.6.3.1), since the "**factory**" command even resets the network configuration

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5.2 Operation via Web Interface

The Web Interface is a comfortable user interface for operating the equipment i.e., the DEV 2x4x/DEV 2x5x is controlled via a computer and a Web Browser as the communication application. Several users (assumed the appropriate number of licenses) can access the DEV 2x4x/DEV 2x5x in parallel. Eight licenses are included in the shipment; please contact DEV Systemtechnik to order additional license keys.

Note:

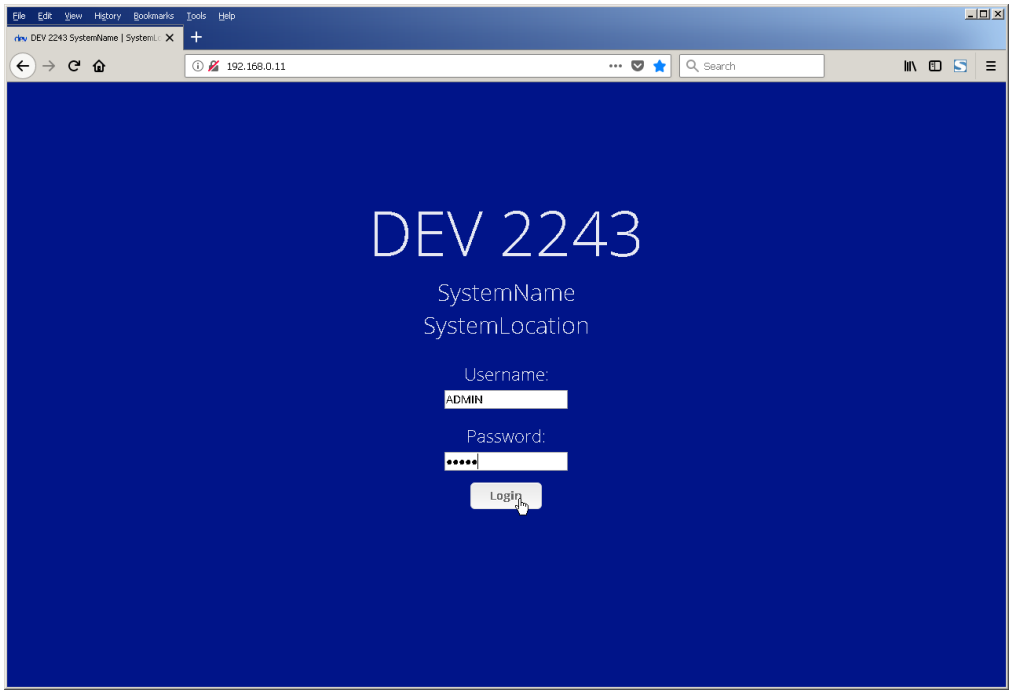
- The screenshots shown in the following were made from the Web Interface of a DEV 2243. Other configurations deviate from the layout but not from the basic functionality of the Web Interface
- The screenshots may vary with respect to date and time; new screenshots are provided for relevant changes only, otherwise they are equivalent to the screenshots of the latest versions of this manual.

5.2.1 Start of the Web Interface

To start the Web Interface, open a web browser (recommended is Mozilla Firefox (version 60 or higher), also Chrome, Opera, or Safari are supported) and type in the URL (IP address) of the DEV 2x4x/DEV 2x5x:

`http:<IP-of-the-equipment>/`

After a moment, the Start Screen of the Web Interface appears:



To log on the first time use the factory default account:

Username: ADMIN
Password: ADMIN

After entering username and password, click on Login.

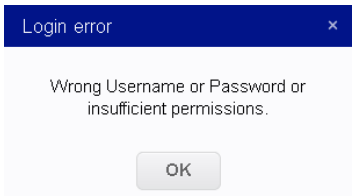
The Web Interface starts loading…

Note:

- Note the header in the Start Screen

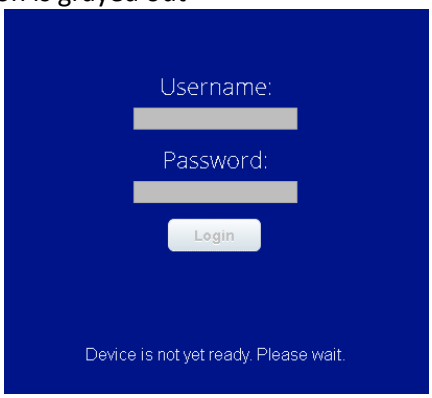
  DEV 2243
  SystemName
  SystemLocation

  The labels "**SystemName**" and "**SystemLocation**" are the SNMP variables 'System Name' and 'System Location'; both can be set within the SNMP Window (please refer to chapter 5.2.6.2.3)
- Both, username and password are case sensitive.
- If the login failed due to a wrong username or password or due to insufficient permissions, a corresponding notification pops up:



- If the Start Screen is not accessible, make sure that the computer accessing the Web Interface is in the same subnet of the common network (chapter 4.4.2.1)
- If the network connection gets lost during a session, an appropriate notification pops up. After confirmation, the web browser is redirected to the Start Screen. If the Web Interface is accessible again, a new authentication is required.
- During power-up or after a reset the text fields for "Username" and "Password" cannot be edited and the Login button is grayed out

or



  If the Web Interface is operational (again), the screen usually automatically changes, thus the input fields can be edited. If not, please use the refresh functionality of your web browser e.g., the function key F5.
- In rare cases, the Web Interface becomes inoperable (mostly after login). A reset of the browser cash (Ctrl+F5) usually solves the problem.
- The Web Interface permits the definition of a number of users for accessing the Web Interface in parallel (in accordance with the number of installed licenses).

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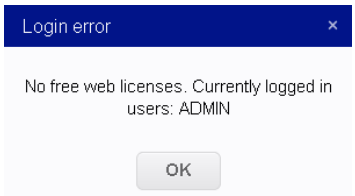
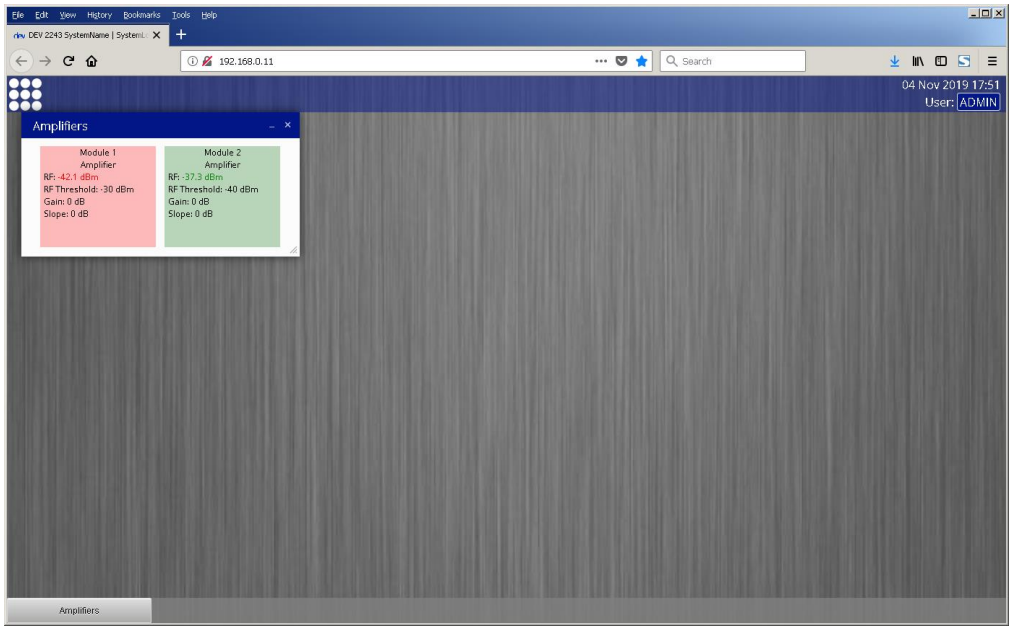
An additional user who tries to log in receives an appropriate notification informing about the logged in users e.g., a notification for equipment with just a single web license installed:



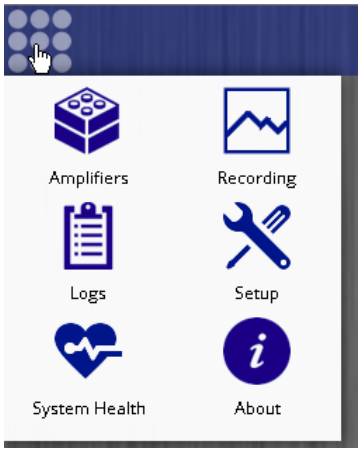
- It is recommended to add user(s) AND to change the password of the factory default account. How this is done is subject of chapters 5.2.6.1.1 & 5.2.2.1

5.2.2 The Web Interface Desktop and Window Handling (,)

After successful authentication and loading, the desktop of the Web Interface shows up with the Amplifiers Window (please refer to chapter 5.2.3) opened:



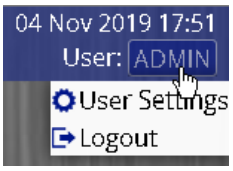
In the top of the window on the left side, the Select Icon provides access to the functionality of the Web Interface. Clicking on shows the icons for the (other) functional windows that are subject of the following chapters.



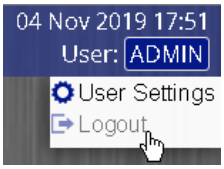
Note:

- The Recording Icon appears only with applied Option 77 (chapter 3.4)

On the right side, the date and time and the currently logged in user (here "ADMIN") are shown. Clicking on the username opens a second menu:



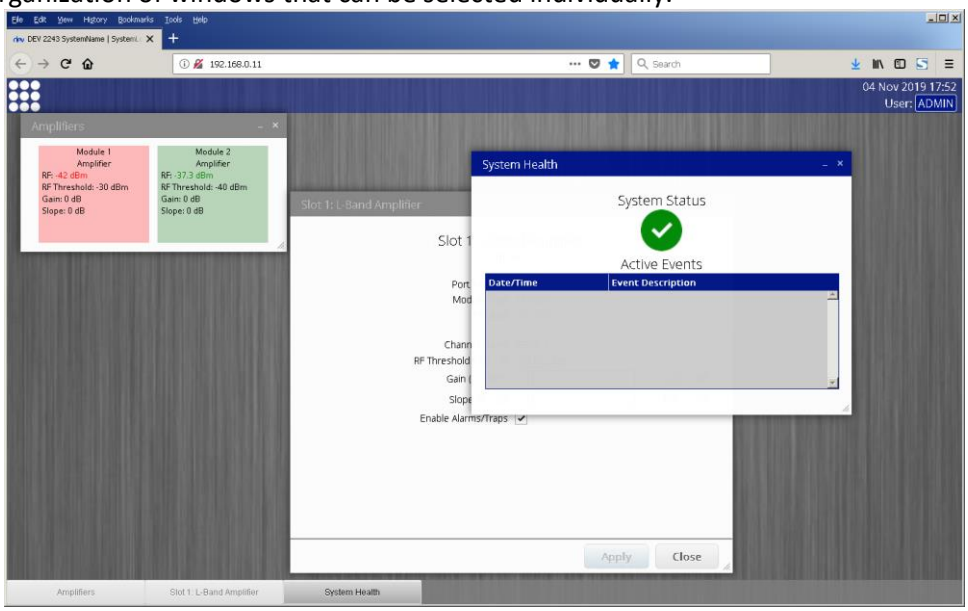
The "User Settings" are explained in chapter 5.2.2.1 and selecting "Logout" terminates the session and returns the Web interface to the Start Screen



Note:

- It is recommended to finish a Web Interface session by logging out since this instantaneously releases the allocated license and enables other users to log in.
- A timeout feature (~20 seconds) is implemented within the Web Interface i.e., a session is automatically terminated (and the allocated license is released) if a user who is currently logged in closes the web browser.
- This means that if the session timeout (default) setting "Off" is applied (please refer to chapter 5.2.6.1.1), a Web Interface session is kept open unless logging out or if the web browser (in other words, the connection to the Web Interface session) is closed.
- It is possible to start a Web Interface session on another computer using the same user account; naturally, this automatically terminates the preceding computer session.

The desktop of the Web Interface permits the flexible presentation and organization of windows that can be selected individually.



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The screenshot above shows the desktop with three windows opened, resized, and repositioned. In addition, the possible resizing of the web browser window itself adds flexibility in the overall presentation of the Web interface.
Some remarks regarding the handling of windows on the Web Interface desktop:

- Usually (additional) windows are opened via the Select Icon and by clicking on a channel representation within the Amplifiers Window (chapter 5.2.3).
- An opened window can be resized with the mouse in the lower right corner of the window

  The window can be dragged to the desired size while pushing the left mouse button.
- An opened window can be positioned with the mouse in the window title bar

  The window can be dragged to the desired position while pushing the left mouse button.
- An opened window can be minimized by clicking with the mouse on the minimize symbol ( ) on the right in the window title bar.
- An opened window can be closed by clicking with the mouse on the close symbol ( ) on the right in the window title bar.
- Additionally, opened windows appear as buttons within the horizontal window bar at the bottom of the desktop

  The active window is highlighted in the window bar, clicking on a different button activates the corresponding window instead.
  If a window is selected in the window bar that was minimized before, will be reopened and becomes the active window. Note that alternatively a (visible) window can become the active window by just clicking somewhere into the window.

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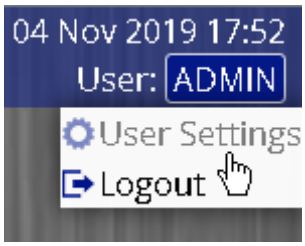
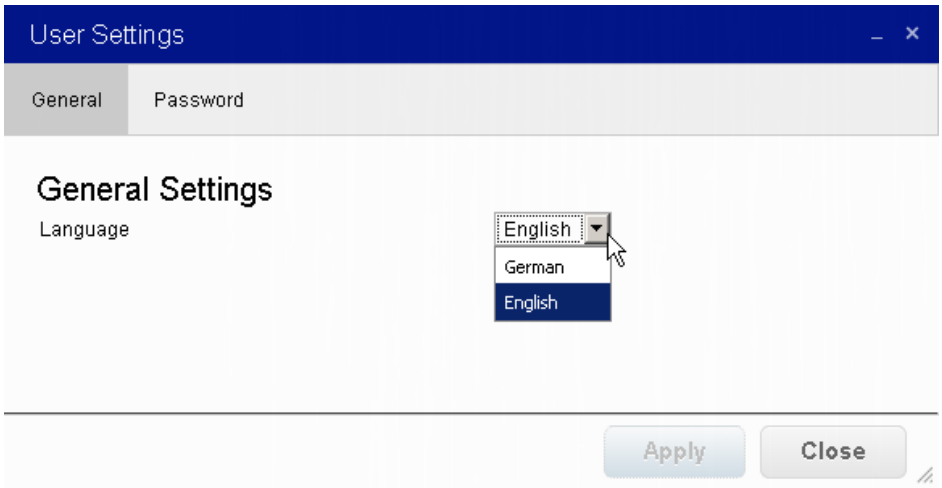
5.2.2.1 User Settings

Selecting "User Settings" on the right side of the desktop…



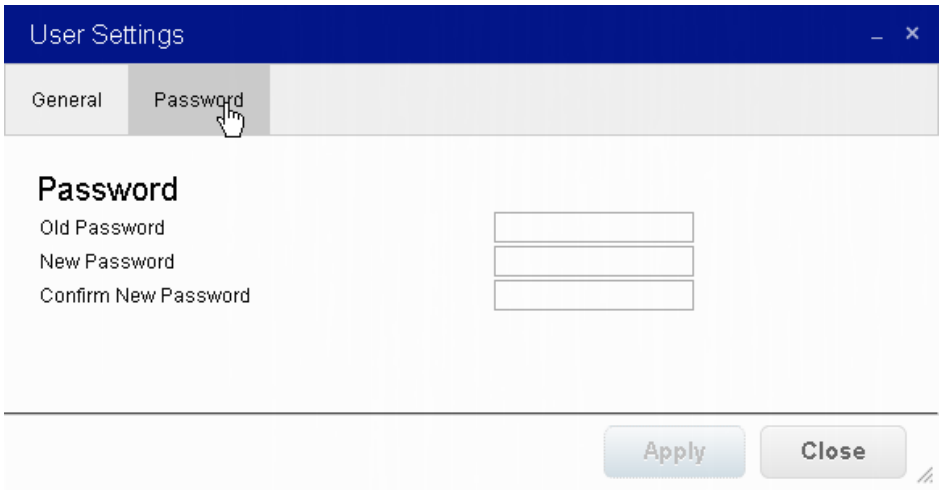
…opens a window consisting of two tabs. The "General" tab permits to change the language by selecting an entry from the drop down list "Language". A click on the button Apply saves the changes made.



Note:

- English and German are the languages supported by the current firmware.

The input fields of the "Password" tab enable the user who is currently logged in to change the password by entering the old password, the new password, and the confirmation of the new password. A confirmation notification pops up if the password has been changed successfully.



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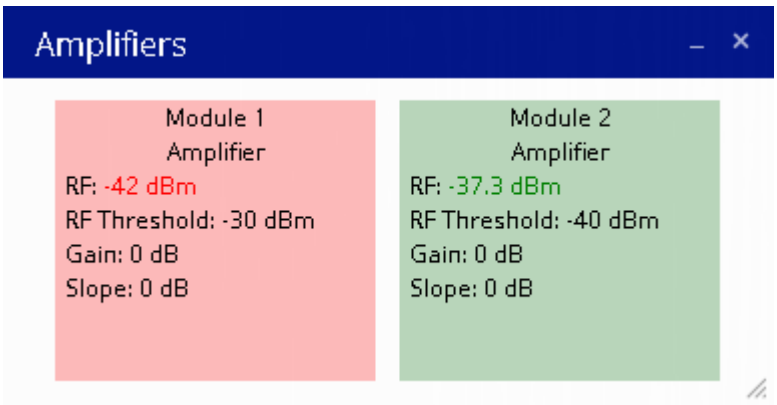
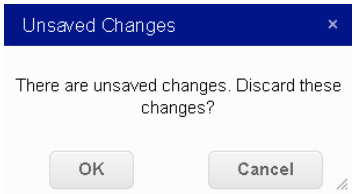
5.2.3 Amplifiers Window



The Amplifiers Window shows a symbolic representation of the amplifier channel(s) providing a quick overview regarding the detected configuration. The health status of the amplifier channel(s) is indicated via appropriate colors for the background and for the values.
Clicking on the representation of an amplifier channel opens the (related) Amplifier Channel Window explained next.

Note:

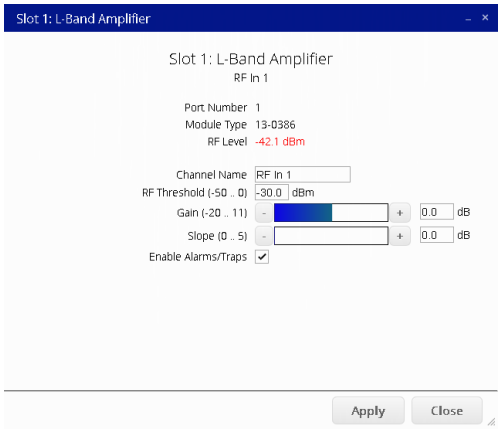
- If changes of settings (which are to be saved explicitly) are applied in an Amplifier Channel Window (or within a window of the Setup Window, please refer to chapter 5.2.6), they are highlighted in yellow. If closing the window without saving the change(s), a corresponding warning notification pops up:

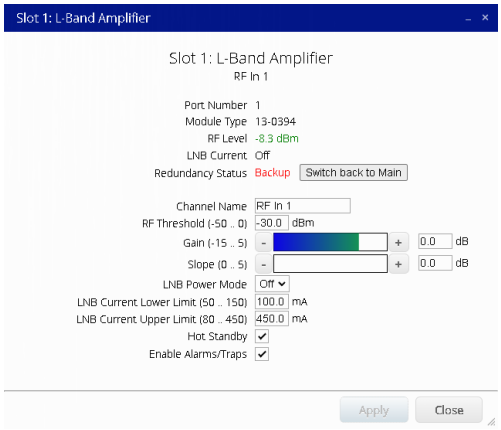


  Cancel closes the notification and permits to save the changes in the window, and OK closes the window discarding the changes.
- Users with "read" permission only have no rights to apply any changes on the settings.

5.2.3.1 Amplifier Channel Window

An Amplifier Channel Window is accessible by clicking on the representation of an amplifier channel in the Amplifiers Window (please refer to chapter 5.2.3).





Below the header lines indicating the location (here "Slot 1"), the amplifier category (here "L-Band Amplifier"), and the channel name (here "RF In 1"), first the "Port Number" is shown (here "1") as a reference to the index of the SNMP .portTable (please refer to chapter 5.3.4.3.3). In the line below, the "Module Type" is listed (here e.g., "13-0386"). Next, the measured "RF Level" is displayed (here e.g., "-42.1 dBm").
Note that the range of the measurement can be exceeded. In this case, a message "Out of Range (Overflow)" or "Out of Range (Undercut)" is displayed in addition to the (faulty) measured level (as shown in the screenshot below).
Then, and for (distribution) amplifiers only, "LNB Current" informs about the LNB power supply status (here "Off"); if activated, the measured current (in [mA]) is presented. Finally, and for redundant amplifiers only (as shown in the right screenshot above), "Redundancy Status" indicates whether the main amplifier path (green "Main") or the redundant amplifier path (red "Backup") is activated.

Note:

- The Switch back to Main button in the Amplifier Channel Window of a redundant amplifier DEV 2x5x appears only if switched to the redundant amplifier path (i.e., red "Backup", as shown in the right screenshot above).
  If e.g., during installation a signal is applied to the input port(s) without the output port(s) properly connected and/or terminated, it can happen that the amplifier switches unintentional to the redundant amplifier path.
  In this case, operate the Switch back to Main button to switch back to the main amplifier path i.e., to re-arm the redundancy functionality.

The following lines in an Amplifier Channel Window permit changes on the parameters of the related amplifier.
Note that after settings have been altered within this section, a click on the Apply button is necessary to save the changes.

- To alter the "Channel Name", enter a new name in the text field.
- The "RF Threshold" for the RF level monitoring functionality can be adjusted within the stated range.
- The "Gain" adjustment can be performed roughly by clicking into the related control bar, in discrete steps by using the - or + buttons, or by directly entering the desired numerical value in the field right to the + button. The

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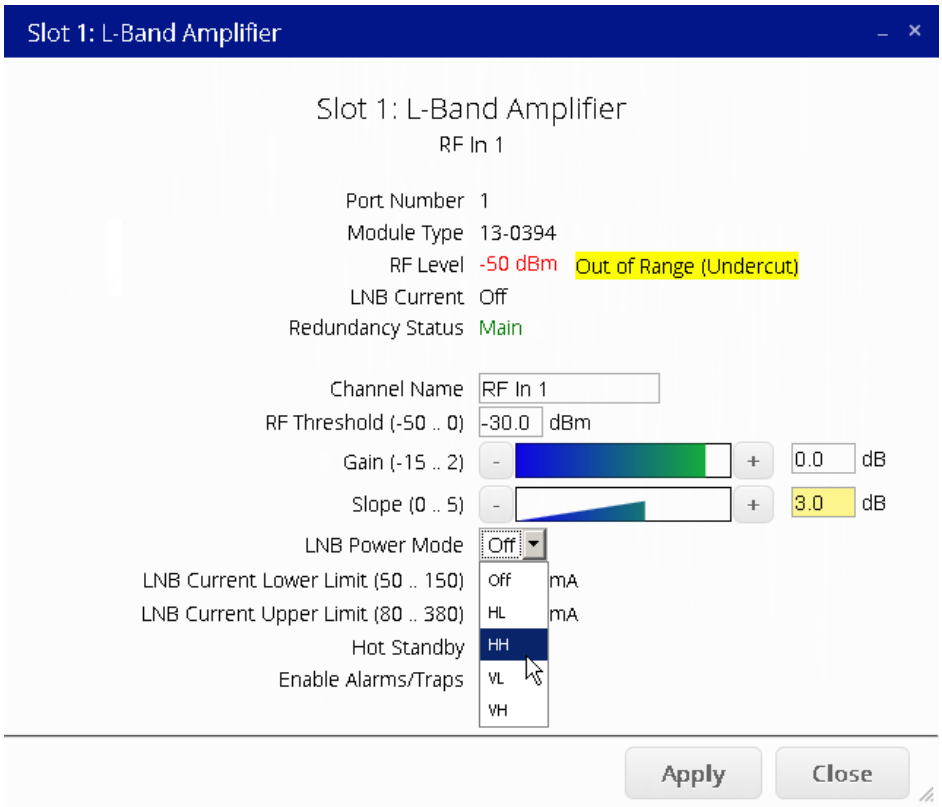
current target setting is displayed via the (blue/green) colored bar and via the numerical value.

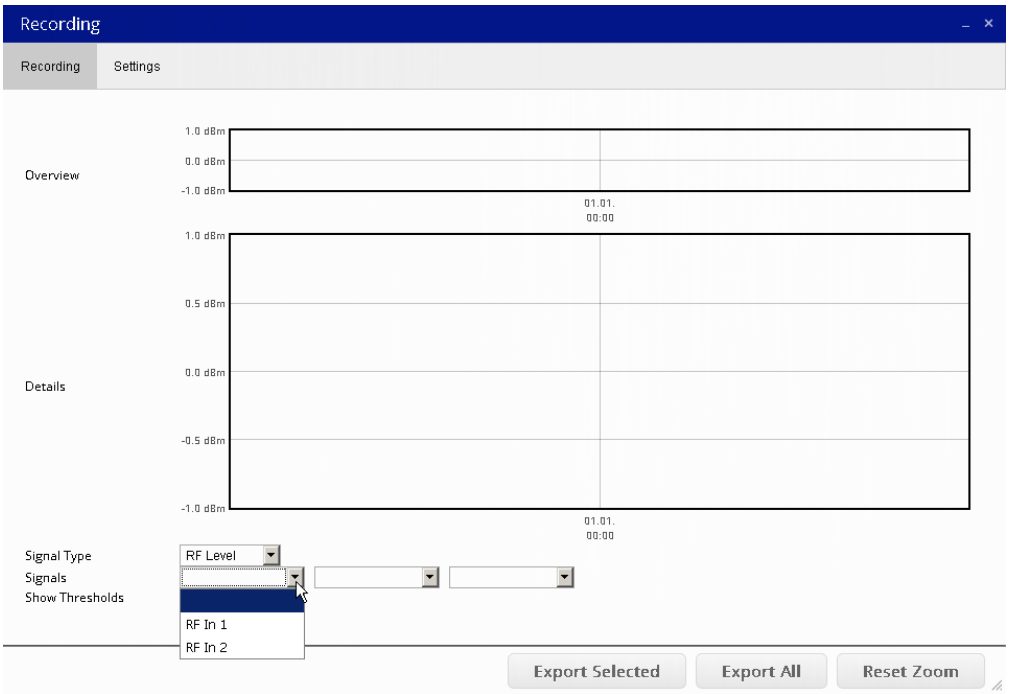
- The "Slope" adjustment can be performed roughly by clicking into the related control bar, in discrete steps by using the - or + buttons, or by directly entering the desired numerical value in the field right to the + button. The current target setting is displayed via the (blue/green) colored bar and via the numerical value.
- For distribution amplifiers only (and as shown in the right screenshot above): If LNB power is required at the RF input of the amplifier, the drop down list "LNB Power Mode" needs to be operated: The default is "Off"; to turn on the LNB power, select one of the polarization/band entries "HL" (horizontal low), "HH" (horizontal high), "VL" (vertical low), or "VH" (vertical high). The entries "LNB Current Lower Limit" and "LNB Current Upper Limit" are for the current monitoring functionality. Their values may be altered within the stated ranges to change the current monitoring interval settings.
  Again, please consider the "LNB power" warnings in chapter 2.3!
- For redundant amplifiers only (and as shown in the right screenshot above): If the main amplifier path is activated ("Redundancy Status" indicates "Main"), the "Hot Standby" check box is used to enable (i.e., hot standby, checked) or to disable (i.e., cold standby, unchecked) the power supply of the redundant amplifier path.
- The "Enable Alarms/Traps" check box is used to enable (default) or to disable events possibly caused by the specific amplifier channel.

Finally, an example for a redundant distribution amplifier without input signal(s) and with parameter changes that are not applied yet:

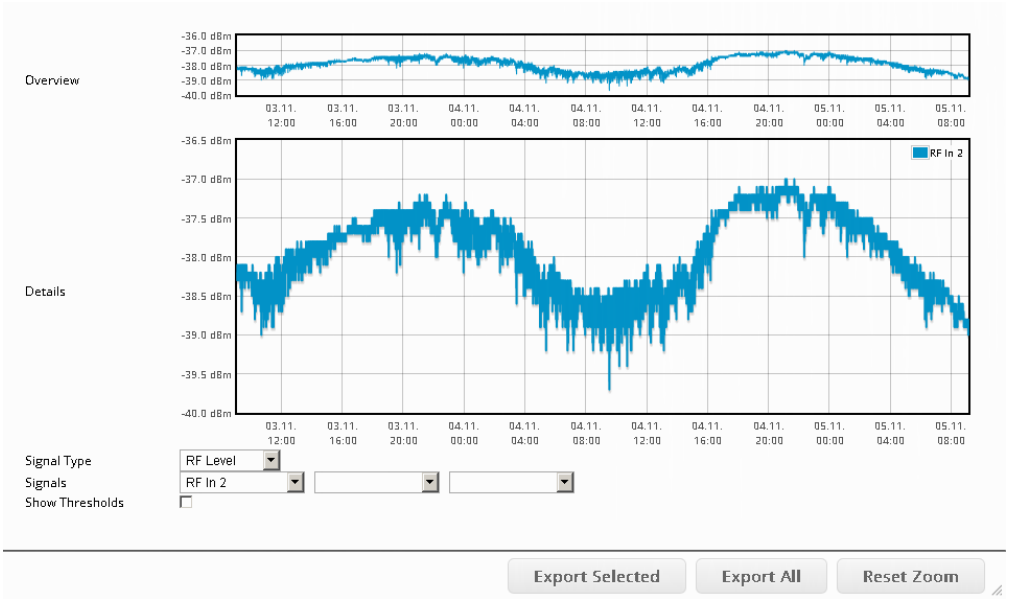


5.2.4 Recording Window

Only with applied Option 77, the Web Interface of the DEV 2x4x/DEV 2x5x provides the Recording Window consisting of the two tabs "Recording" and "Settings".
Some data of the (each) amplifier is continuously captured in discrete intervals. The "Recording" tab of the Recording Window permits the visualization and the external storage of this data. After opening the Recording Window for the first time in a Web Interface session, two empty diagrams labeled "**Overview**" and "**Details**" show up in the initially activated "Recording" tab:



After selecting "RF Level" or -for distribution amplifiers only- "LNB Power" (for LNB current monitoring data) from the drop down list "**Signal Type**" and after selecting 1…3 "**Signals**" via the drop down lists, the recorded data is shown in both diagrams:



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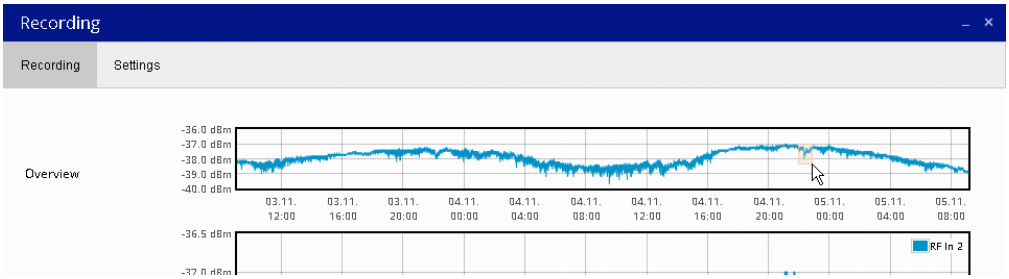
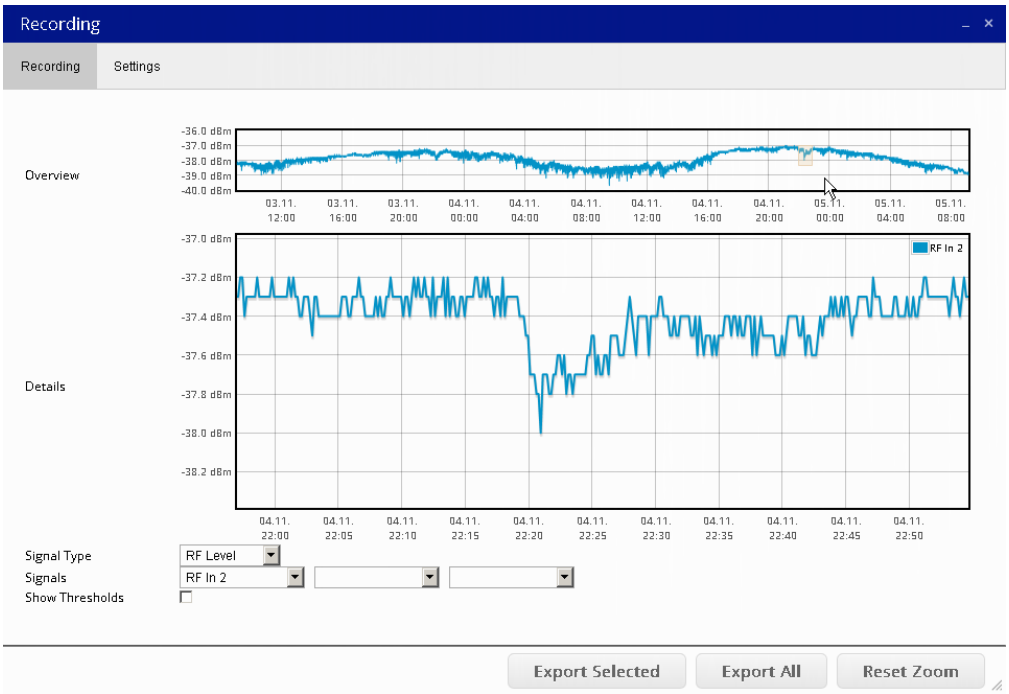
Note:

- The recording interval and thus the recording duration can be altered, please refer to the description of the "Settings" tab at the end of this chapter
  The data representation initially comprises the recorded data for the selected items i.e., it is stretched if the full recording duration is not reached yet, and the graphs indicate via gaps if the equipment was turned off for a span of time.
- The checkbox labeled "**Show Thresholds**" enables the display of the related present threshold(s) for the selected signal(s). If checked, each threshold is shown as a separate horizontal line with an individual label.

It is possible to magnify the data by drawing a zoom window in either (!) diagram i.e., by clicking and holding the left mouse button while pulling the mouse over the interesting area:



Directly after releasing the left mouse button the zoomed area appears in the "**Details**" diagram while the zoomed area is marked in the "**Overview**" diagram:



Three buttons are located at the bottom of the "Recording" tab.
If the recorded data was zoomed as described above, the button Reset Zoom resets the data representation to the initial state i.e., for the selected signals all recorded data of the specified signal type is displayed in both diagrams.
The other two buttons provide the option to store the captured data externally (without the possibly displayed thresholds). Export Selected stores the recorded

data of the currently selected signals and the button Export All stores all recorded data i.e., the available data of all signals. After operating one of the buttons, a browser specific window pops up and suggests the storage of the data as "recording.csv" and as "recording_all.csv", respectively. The csv (comma-separated values) file format permits the data import e.g., into a MS Excel worksheet:
The csv-file is a text file modeling a table. Each line is interpreted as row containing column information, here with the columns delimited by a semicolon (";") character as the column separator.
The number of columns depends on the export method (Export Selected vs. Export All) and how many signals are currently presented if the Export Selected button is chosen. In any case, each signal consists of a "`Time`" and of a "`Values`" column; the number of lines depends on the recording duration (over 17000 lines e.g., within two days with the default ten seconds recording interval).
The first line of the csv-file is the header line providing the column information. The first entry is assigned to the first signal (e.g., "`RF In 2`", indicating the channel name of the first signal as described in chapter 5.2.3.1), concatenated with the "**Signal Type**" (i.e., either "`RF Level`" or "`LNB Power`") and with "`Time`". After the semicolon, the same entry is listed with "`Values`" instead of "`Time`". This sequence is repeated for all columns that are part of the csv-file.
The actual data starts in the second line of the csv-file. Note that the time information is stated in standard MS Excel format i.e., the integral part of the number defines the date (starting from 1 being the 01-Jan-1900 e.g., 43772 denotes the date 03-Nov-2019) and the fractional part defines the time within the date (1/(24 * 60 * 60) = 00:00:01 e.g., .375 = 09:00:00).
In the following just the first few lines of a csv-file are listed, which was created by exporting the data shown in this chapter via the button Export Selected:

```
RF In 2 RF Level Time;RF In 2 RF Level Values;
43772.379155092596;-38.2;
43772.379270833335;-38.2;
43772.37938657407;-38.2;
43772.37950231481;-38.2;
43772.37961805555;-38.2;
43772.3797337963;-38.3;
43772.379849537036;-38.2;
```

After import and formatting the "`Time`" columns to the format "DD-MMM-YYYY, hh:mm:ss" this results in:

RF In 2 RF Level Time	RF In 2 RF Level Values
03-Nov-2019, 09:05:59	-38.2
03-Nov-2019, 09:06:09	-38.2
03-Nov-2019, 09:06:19	-38.2
03-Nov-2019, 09:06:29	-38.2
03-Nov-2019, 09:06:39	-38.2
03-Nov-2019, 09:06:49	-38.3
03-Nov-2019, 09:06:59	-38.2

Note:

- The decimal separator in the csv-file is the point character ("."), this has to be considered during the data import of the csv-file e.g., into an Excel worksheet.
- For the data import into an empty MS Excel worksheet select the tab "Data" and there the icon "From Text". Then locate the csv-file and follow the instructions of the text import wizard.

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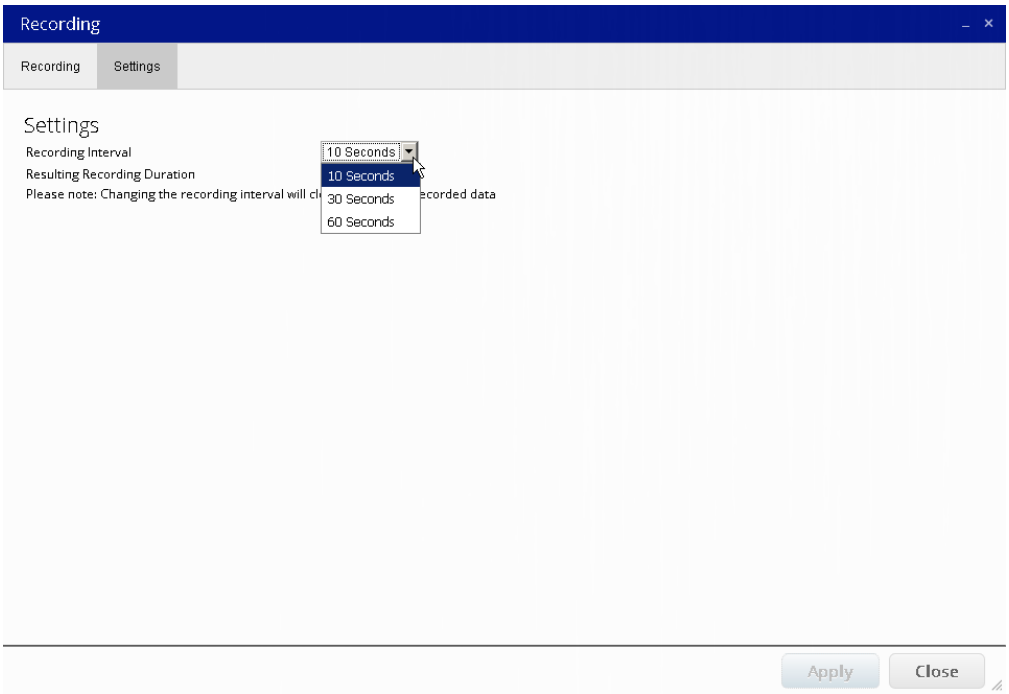
- The correct time information is dependent on the proper configuration of a timeserver and of the timezone. These settings can be applied via the Date & Time Window (chapter 5.2.6.2.2). If no timeserver is configured, the recording of the data starts at 01-Jan-2000, 00:00:00; after a device reset the time is reset to 01-Jan-2000, 00:00:00 as well, but the recording of the data is continued with the time of the last sample.
  If a timeserver was specified that is not available anymore after a device reset, the recording of the data is continued with the time of the last sample as well. Otherwise, if the timeserver is available after a device reset, the recording of the data is continued with the correct time i.e., this causes a time gap within the data.



Finally, the "Settings" tab of the Recording Window permits the configuration of the interval for the signal recording functionality.
Via the "Recording Interval" drop down list the interval for the regular data capture can be selected. After an entry was picked, the "Resulting Recording Duration" is displayed in the line below. The following choices are available:

Recording Interval	Resulting Recording Duration
10 Seconds	2 Days
30 Seconds	6 Days
60 Seconds	12 Days

Note:

- After clicking on the Apply button with a different recording interval selected than the currently applied setting all stored recording data is deleted instantaneously; the recording starts with the new recording interval setting.
- The default recording interval setting is 10 seconds; after a Restore Factory Defaults (chapters 5.1.12 & 5.2.6.3.1), this setting is applied as well.

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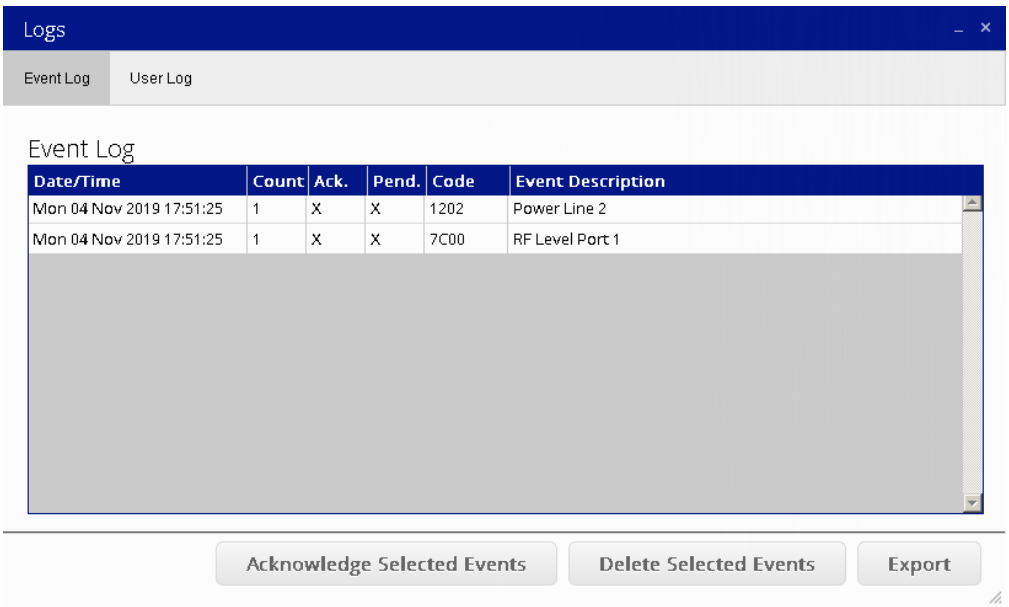
5.2.5 Logs Window



The Logs Window provides the two tabs "Event Log" and "User Log". With the "Event Log" tab selected, a table with the detected events is displayed as shown above.
The different events are listed line-by-line i.e., each event carries a "**Date/Time**" stamp (in the format "DDD DD MMM YYYY hh:mm:ss", here "Mon 04 Nov 2019 17:51:25") informing about the last occurrence of that event. The second column is a "**Count**"er reporting the number of occurrences of that event (here "1"). The third column denotes whether that event is "**Ack.**"nowledged ("x") or not (" ") (here "x" meaning acknowledged). The fourth column shows whether that event is "**Pend.**"ing ("x") or not (" ") (here "x" i.e., pending). The fifth column states the error "**Code**" in hexadecimal notation (here e.g., "1202"). The last column reports the related "**Event Description**" (here e.g., "Power Line 2"), please refer to chapter 6.1.

Below the table, the three buttons Acknowledge Selected Events, Delete Selected Events, and Export are located: An event can be acknowledged by highlighting (i.e., clicking) the related line and by operating the Acknowledge Selected Events button afterwards. The intention of the event acknowledgement is to provide means to hide a pending event in the System Health Window (chapter 5.2.7). If a confirmed event occurs again, the counter in the table is increased, the acknowledgement is reset (i.e., the "x" disappears), and the event shows up again in the System Health Window.

An acknowledged and not pending event can be deleted from the table by highlighting the corresponding line, followed by clicking on Delete Selected Events.
If a deleted event occurs again, a line added at the beginning of the table.

After operating the Export button, a browser specific window pops up and proposes the storage of the table as "eventlog.csv". The .csv (comma separated values) file contains a line per log entry; each of the eight columns of a log entry is delimited by a semicolon (";") character as the column separator.

Note:

- Users with "read" permission (chapter 5.2.6.1.1) only have no rights to acknowledge or to delete event entries.

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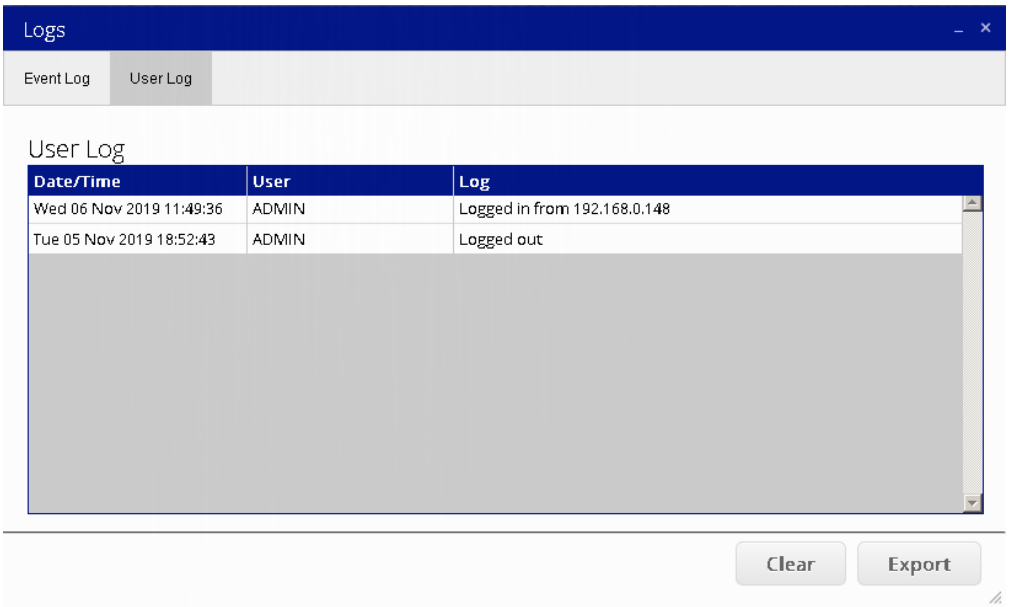
- It is possible to acknowledge and to delete more than one entry at a time by highlighting more events using the <Ctrl> or the <Shift> key of the keyboard in combination with mouse click(s).
- If an event is still pending or not acknowledged, it cannot be deleted.

If the entry "User Log" tab is selected in the Logs Window, a table providing user activity data appears:



The data is listed line-by-line i.e., each log entry carries a "**Date/Time**" stamp (in the format: "DDD DD MMM YYYY hh:mm:ss", here e.g., "Wed 06 Nov 2019 11:49:36"). The second column shows the "**User**" (here "ADMIN"). The "**Log**" column reports the logged activity (here e.g., "Logged in from 192.168.0.148").

Below the table, the Clear button and the Export button are located: With the Clear button, the table is cleared (without confirmation) and after operating the Export button, a browser specific window pops up and proposes the storage of the table as "userlog.csv". The .csv (comma separated values) file contains a line per log entry; each of the three columns of a log entry is delimited by a semicolon (";") character as the column separator.

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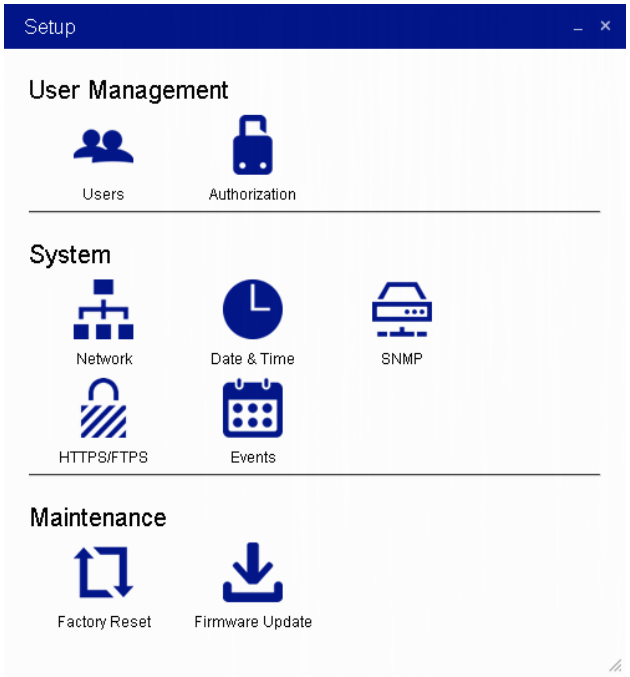
5.2.6 Setup Window

The Setup Window is divided in three sections structuring the different windows described in the following.



Note:

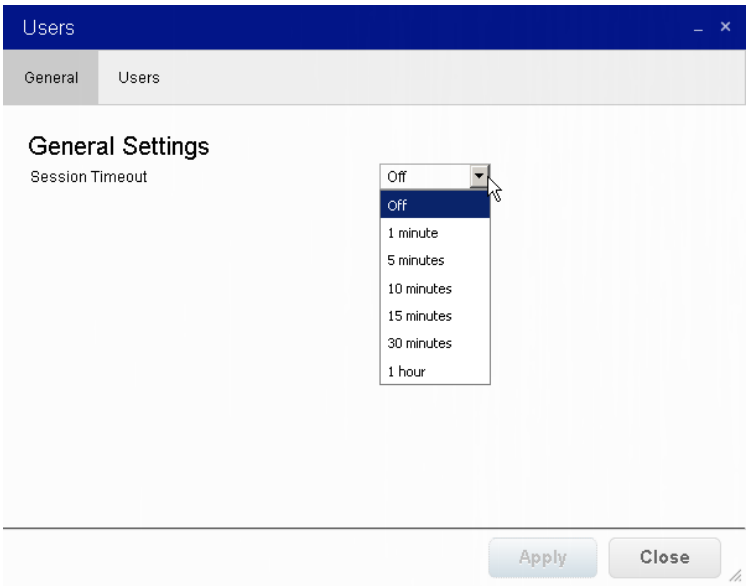
- Only users with "admin" permission have rights to access the Setup Window

5.2.6.1 User Management

The "**User Management**" section in the Setup Window provides access to the Users Window and to the Authorization Window.

5.2.6.1.1 Users Window

The Users Window provides the two tabs "General" and "Users".



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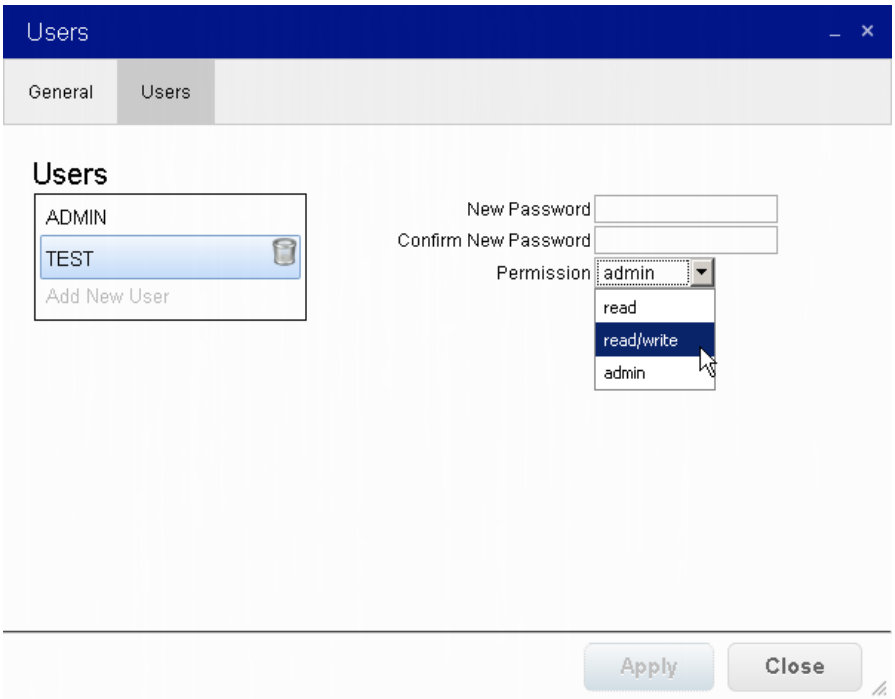
With the "General" tab selected (default) in the Users Window, a session timeout for Web Interface sessions can be set. As stated in chapter 5.2.2, the default setting is "Off". Selecting any other value from the drop down list "Session Timeout" and saving it via the Apply button applies a timeout for all Web Interface sessions i.e., the return to the Start Screen after the session being idle for the selected duration.



With the "Users" tab selected in the Users Window, users can be added or deleted, and the permission can be modified if the standard "Local" authorization method is activated, please refer to chapter 5.2.6.1.2.
In the screenshot above, a user named "TEST" with "admin" permission is in the process to be restricted to "read/write" permission. After a click on the Apply button, this user will then have all rights but no access to the Setup Window. A user with "read" permission additionally is not able to apply any changes within all windows accessible via the Select Icon of the Web Interface.
To add a new user, click on the item "Add New User" and enter a name in the appearing text field and click again. By default, this new user has "read" permission and no password is set for this user.
By activating an existing user, the permission category can be altered and a password can be set for this user. A click on the Apply button saves the changes made. If the password has been changed, a confirmation window pops up.
To delete a user, click on the (trash bin) icon right to the highlighted user entry, a window pops up for confirmation.

Note:

- As already mentioned, usernames and passwords are case sensitive.
- If the active authorization method is "RADIUS" or "LDAP" (chapter 5.2.6.1.2), only the local (not necessarily logged in) "ADMIN"-password can be changed here.
- It is not possible to delete the "ADMIN" user, but it is recommended to change the default password.

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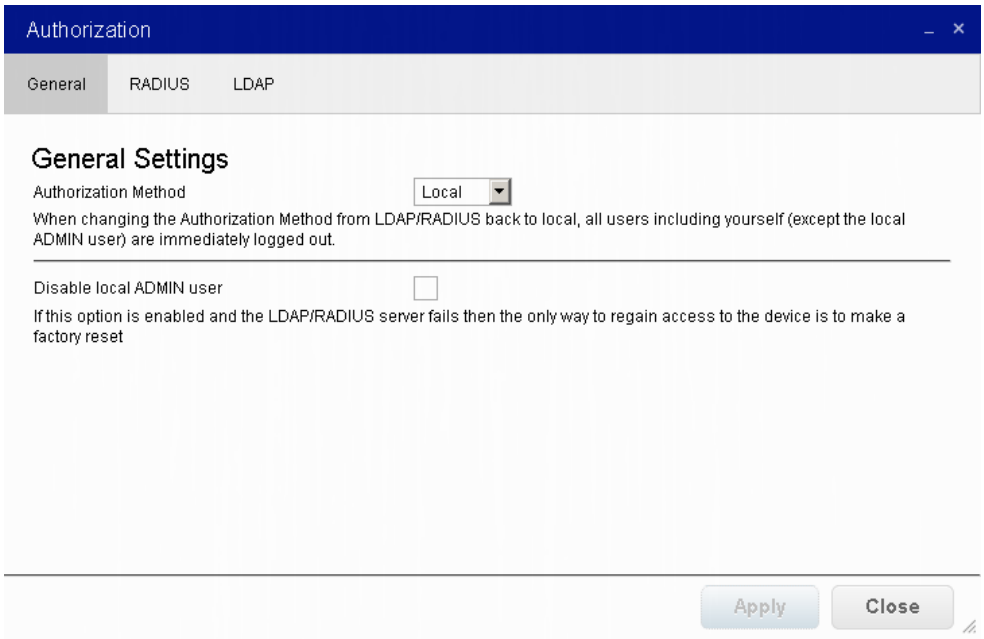
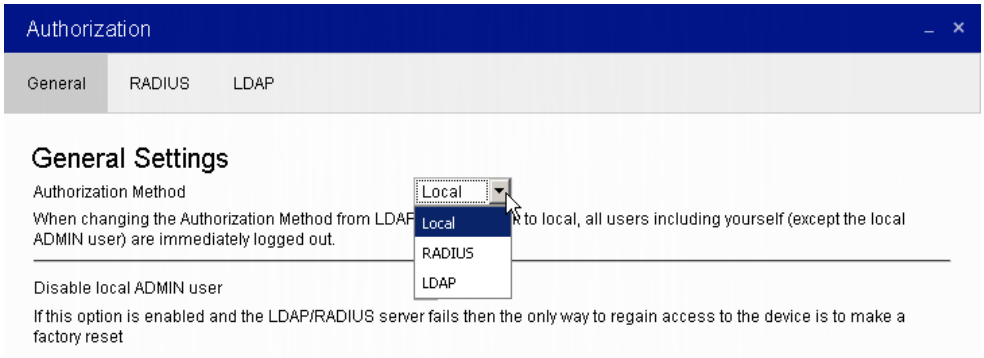
5.2.6.1.2 Authorization Window

The Authorization Window provides the three tabs "General", "RADIUS", and "LDAP".



In the "General" tab, the drop down list "Authorization Method" provides three entries defining the user community that can access the Web Interface:

- "Local" This is the standard authorization method. The users are defined and managed via the User Window (please refer to chapter 5.2.6.1.1)
- "RADIUS" The user community is defined and managed by RADIUS server(s)
- "LDAP" The user community is defined and managed by an LDAP server



Note:

- It is recommended to check the related settings in the "RADIUS" (or "LDAP") tab before the activation of the RADIUS (or LDAP) authorization method via the Apply button.
- Please consider that "When changing the authorization method from LDAP/RADIUS back to local, all users including yourself (except the local ADMIN user) are immediately logged out."

Additionally, in the "General" tab, the local "ADMIN" user can be disabled via a checkbox, please note:

"If this option is enabled and the LDAP/RADIUS server fails then the only way to regain access to the device is to make a factory reset"

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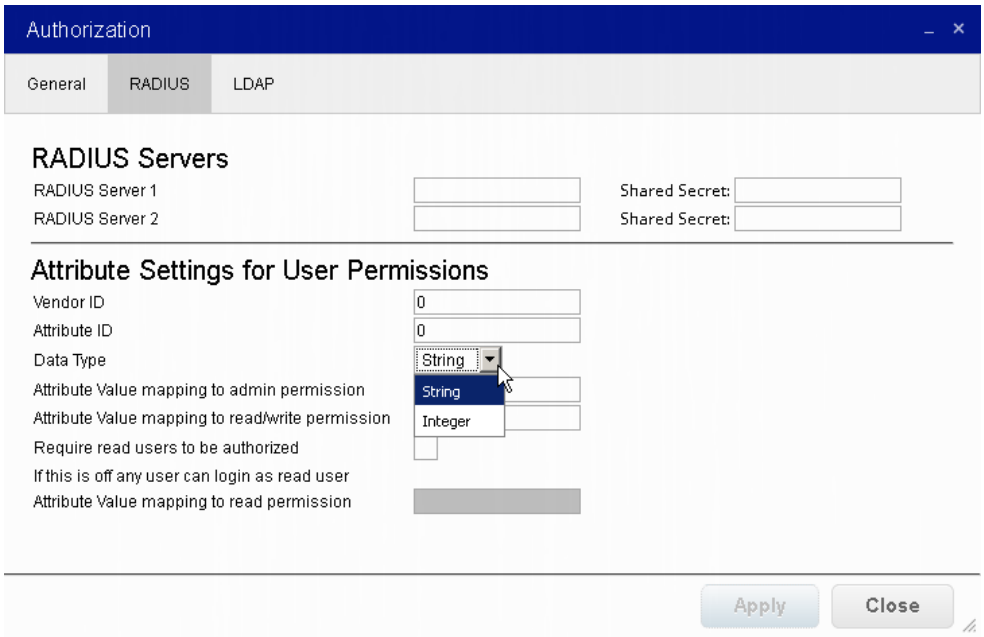
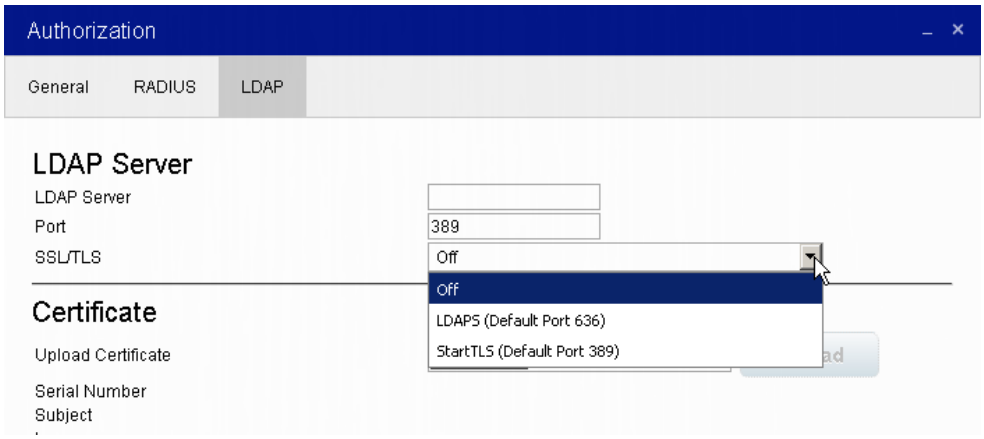


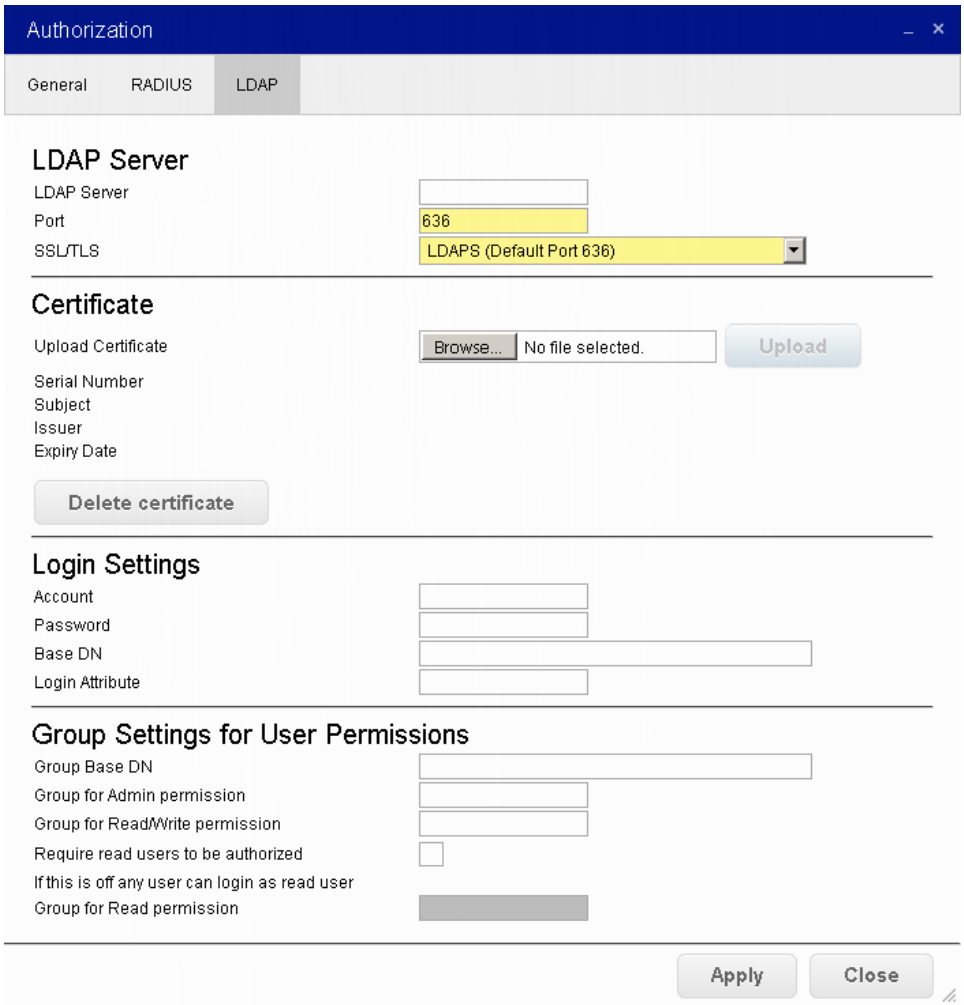
With the "RADIUS" tab activated, the section labeled "**RADIUS Servers**" is for the specification of one or two RADIUS servers via IP address –the RADIUS port number 1812 is added automatically- and via the (related) "Shared Secret". The symbol right to the RADIUS server IP address field(s) indicates whether a saved (!) entry is an available network resource (✓ green check mark) or not (⚠ red warning sign).

The fields in the section "**Attribute Settings for User Permissions**" are required to grant RADIUS users the different permissions for accessing the Web Interface. The syntax for these fields is RADIUS specific; please contact DEV Systemtechnik for more information on this subject.

A click on the Apply button saves the changes made.



With the "LDAP" tab activated, the section labeled "**LDAP Server**" is for the specification of an LDAP server e.g., via the IP address. After saving using the Apply button, the symbol right to the LDAP server field indicates whether the saved (!) entry is an available network resource (✓ green check mark) or not (⚠ red warning sign). Next, enter the "Port" number (default entry is "389"), and possibly select "SSL/TLS" via the related drop down list.



If "LDAPS" (or "StartTLS") is selected, the "Port" number is adjusted automatically to "636" (or to "389") which can be altered before saving. The two methods do require a valid certificate in .pem format. Thus, the "Upload Certificate" input field in the "**Certificate**" section permits the insertion of a filename including the path where the file resides. Alternatively, the Browse... button can be used to locate the file. After the selection of the file, click on the Upload button to activate the file transfer to the DEV 2x4x/DEV 2x5x.

A stored certificate can be deleted by operating the button Delete certificate plus the confirmation via the appearing pop up window.
For the "Account" and "Password" entries in the "**Login Settings**" section, a service LDAP user is to be entered who has access to the directory structure of the LDAP server. "Base DN" defines the distinguished name (DN) for accessing the naming structure of the LDAP server e.g., uid=user,ou=people,dc=example,dc=com and "Login Attribute" may be used to add the LDAP login attribute e.g., sAMAccountName.
The fields in the section "Group Settings for User Permissions" are required to grant LDAP users the different permissions for accessing the Web Interface. The syntax for these fields is LDAP specific; please contact DEV Systemtechnik for more information on this subject.

A click on the Apply button saves the changes made.

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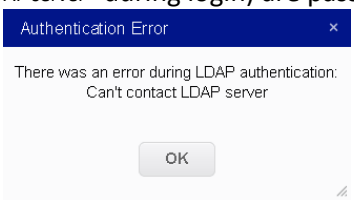
Note:

- If the active authorization method is "RADIUS" (or "LDAP"), all approved users of the RADIUS (or of the LDAP) server may access the Web Interface using their RADIUS (or their LDAP) authentication.
- If using LDAP as authorization method, LDAP specific messages (here e.g., "Can't contact LDAP server" during login) are passed via pop up window.

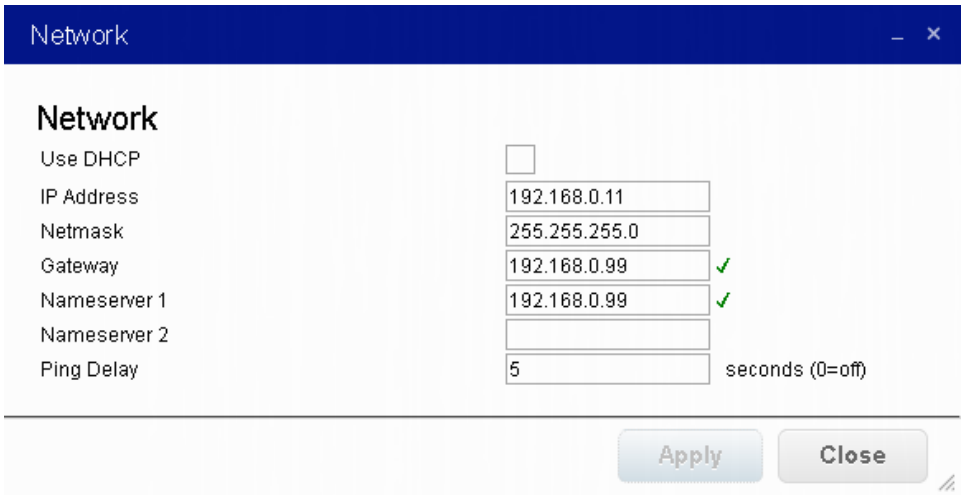


- If the configured RADIUS or LDAP server is not available during login, only the (enabled) local ADMIN account can be used for accessing the Web Interface.

5.2.6.2 System

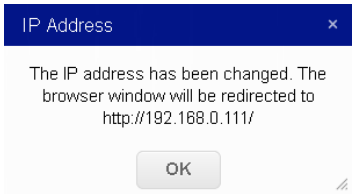
The "**System**" section in the Setup Window provides access to the Network Window, to the Date & Time Window, to the SNMP Window, to the HTTPS/FTPS Window, and to the Events Window.

5.2.6.2.1 Network Window



For the network setup, either "Use DHCP" is to be checked and to be applied to obtain the network parameters automatically, or at least the fields "IP Address" and "Netmask" have to contain valid entries. "Gateway", "Nameserver 1", and "Nameserver 2" are to be specified by their IP addresses – here "192.168.0.99" for "Gateway" & "Nameserver 1" -; the "Nameserver" fields are used for the name resolution of the LDAP Server (Authorization Window, please refer to chapter 5.2.6.1.2) and for the name resolution of the timeserver(s) (Date & Time Window, chapter 5.2.6.2.2).
By default, every five seconds the Web Interface checks the communication to the DEV 2x4x/DEV 2x5x via ICMP Echo Requests (better known as "ping"). To alter the produced network traffic the "Ping Delay" can be used; a higher value reduces the traffic, entering and saving "0" turns off the pinging. The DEV 2x4x/DEV 2x5x has to be restarted (chapter 5.2.6.3.1) that changes on this setting become effective.
A click on the Apply button saves the changes made.

If the IP address has been changed, a confirmation window pops up informing about the redirection of the browser to the new address e.g.,



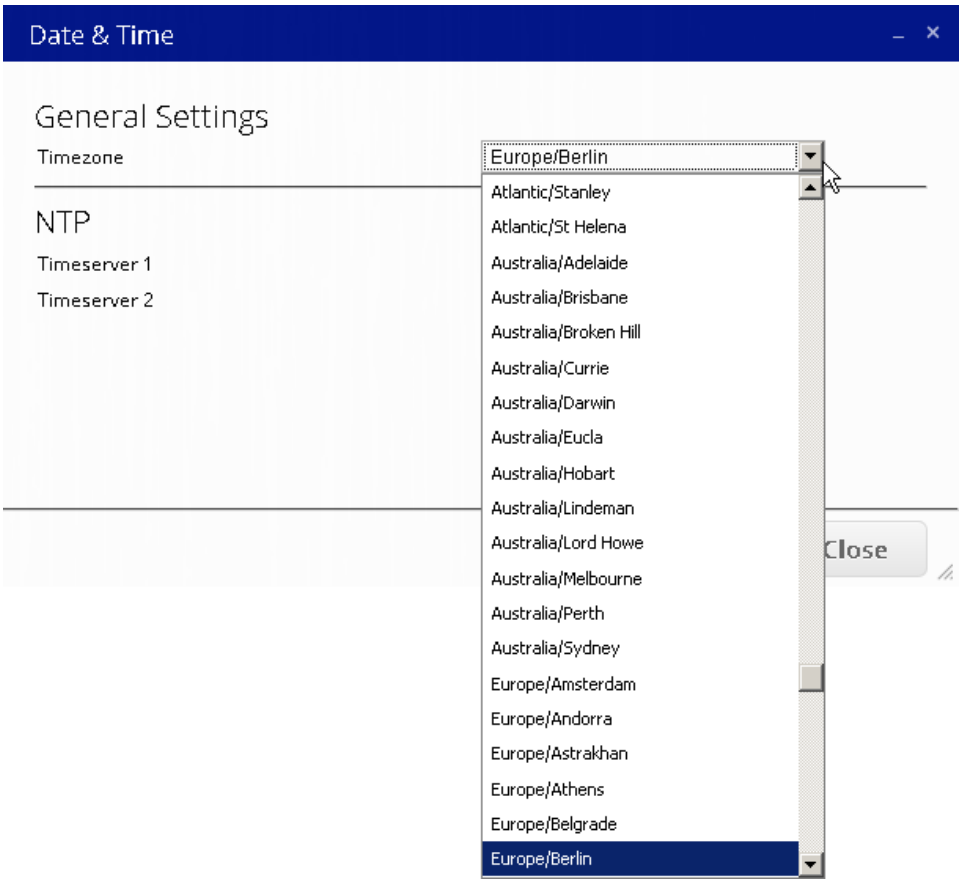
Note:

- If "Use DHCP" is selected and applied, the network parameters are automatically obtained from the local DHCP server. Ask your network administrator for the details on these network parameters to figure out, which parameters might be used by the equipment. Or contact DEV Systemtechnik for the MS Windows application "DEV Device Discovery and Configuration Tool"
- The symbol right to the IP address field(s) indicates whether a saved entry is an available network resource (✓ green check mark) or not (⚠ red warning sign).

5.2.6.2.2 Date & Time Window

The Date & Time Window is used to specify the timezone and one or two timeservers.

To specify the timezone select an entry from the drop down list "Timezone" in the "**General Settings**" section:



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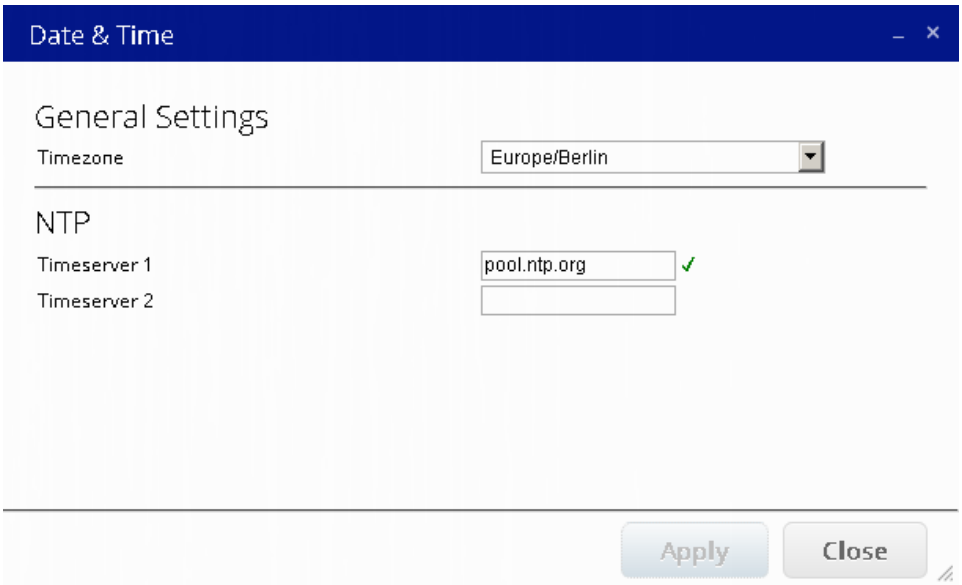


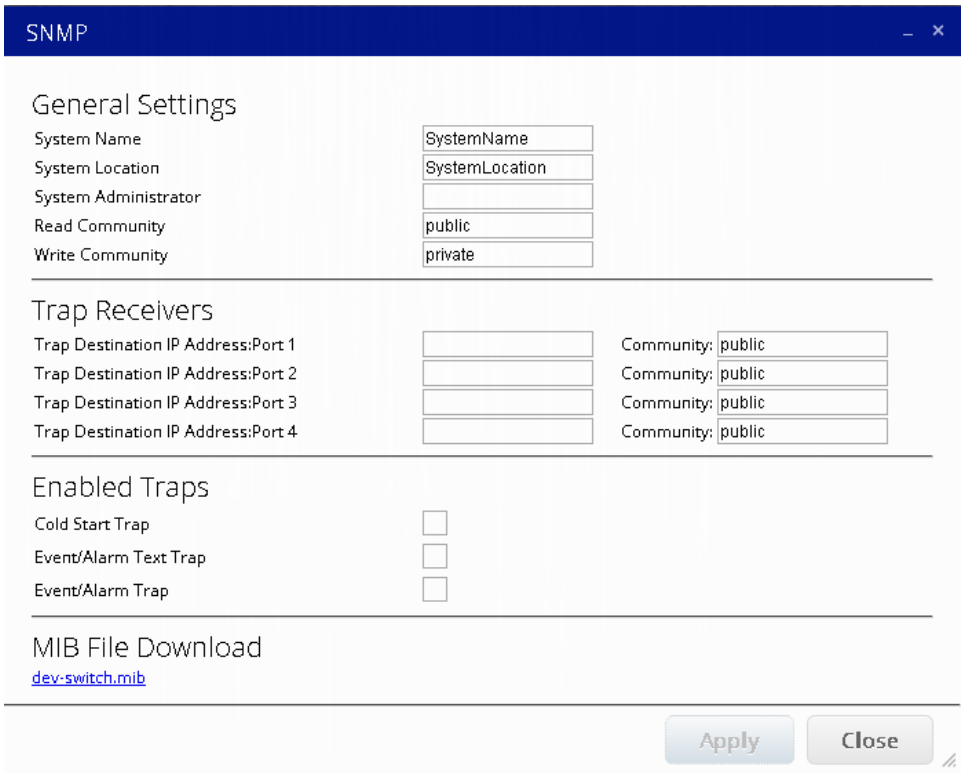
In the "NTP" section, one or two IP addresses or full-qualified domain names (here the single entry "pool.ntp.org" - this requires a nameserver definition; please refer to chapter 5.2.6.2.1) - can be entered in the "Timeserver" fields.

A click on the Apply button saves the changes made.

Note:

- The symbol right to the timeserver field(s) indicates whether a saved entry is an available network resource (✓ green check mark) or not (⚠ red warning sign).

5.2.6.2.3 SNMP Window

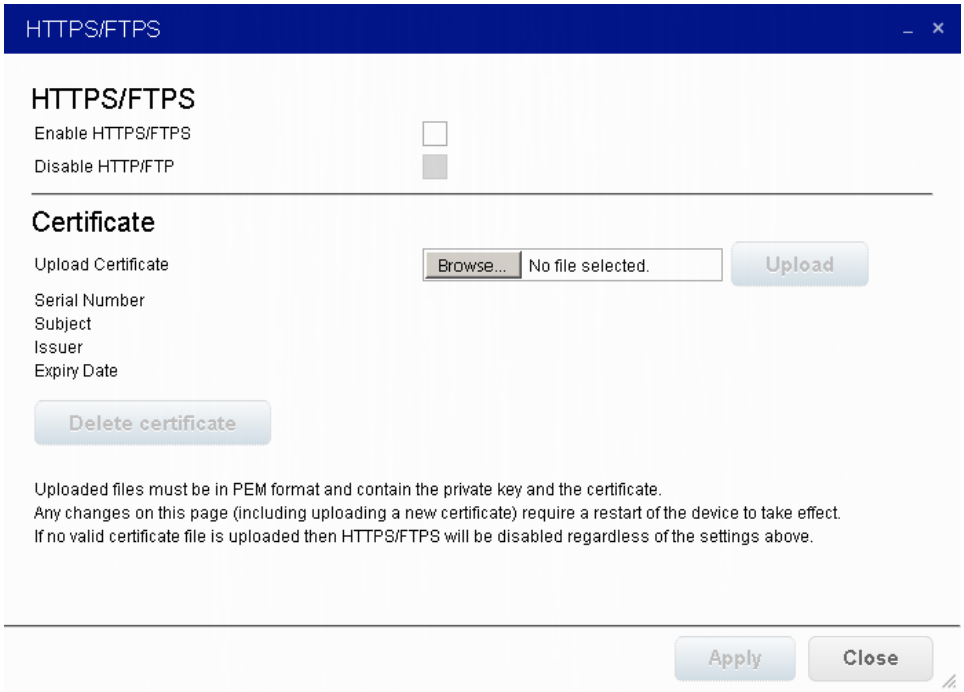


In the "**General Settings**" section of the SNMP Window, the settings for "System Name" and "System Location" can be modified. A "System Administrator" can be defined and the passwords for the "Read Community" and for the "Write Community" can be set.
In the "**Trap Receivers**" section, up to four Trap Destination IP Addresses can be defined via IP address and port in the format <ip-address>{:<port>} e.g., 192.168.0.148:121. If <port> is not specified, the standard SNMP trap port 162 is assumed. The symbols right to the trap destination IP address field(s) indicates whether a saved (!) entry is an available network resource (✓ green check mark) or not (⚠ red warning sign).
The community for each of the four entries can be altered from the default "public" as well.
The check boxes in the section "Enabled Traps" provide access to the SNMP .trapEnableMask (chapter 5.3.4.4.1) via Web Interface.
Finally, a click on the link in the section "**MIB File Download**" initiates the download of the MIB.
A click on the Apply button saves the changes made.

Note:

- The strings "System Name" and "System Location" are part of standard SNMP functionality and are limited to 40 characters, each. The number of visible characters in the text fields is limited by the field size.
- As mentioned, the strings "System Name" and "System Location" are also displayed in the Start Screen of the Web Interface (chapter 5.2.1)

5.2.6.2.4 HTTPS/FTPS Window



The HTTPS/FTPS Window provides means to enable the HTTPS/FTPS functionality and possibly to deactivate the HTTP/FTP functionality.

As stated in the "**Certificate**" section, the HTTPS/FTPS functionality can be activated with a valid certificate in .pem format, only. Thus, it is recommended to upload a certificate file in the first step.

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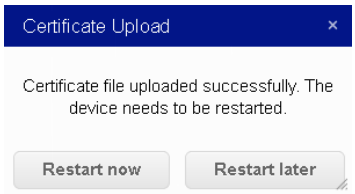
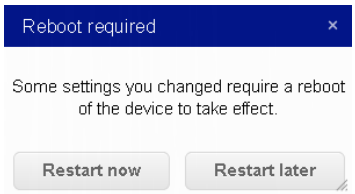
The "Upload Certificate" input field permits the insertion of a filename including the path where the file resides. Alternatively, the Browse... button can be used to locate the file. After the selection of the file, click on the Upload button to activate the file transfer to the DEV 2x4x/DEV 2x5x. The successful upload is indicated via a pop up window



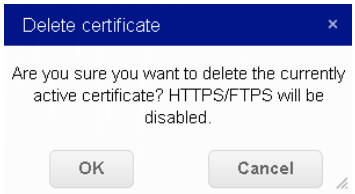
It is now recommended to click on Restart later to check "Enable HTTPS/FTPS" and possibly "Disable HTTP/FTP". A click on the Apply button initiates another pop up window:



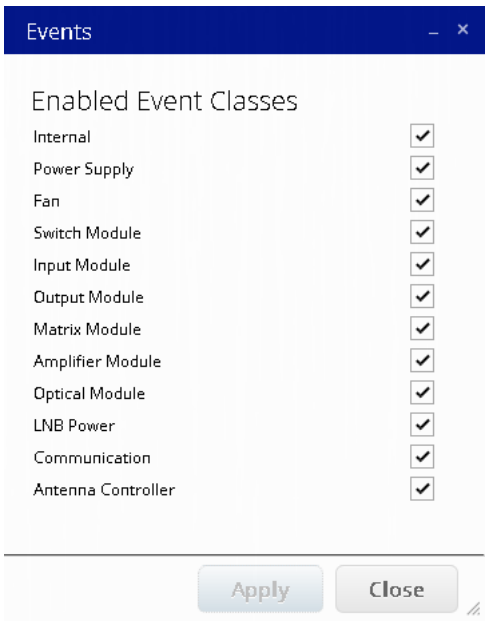
Now the equipment may be restarted by clicking on the Restart now button.

Note:

- HTTPS enabled means that the secure hypertext transfer protocol can be used for the access to the Web Interface in addition. The Web Interface can be accessed exclusively via HTTPS if "Disable HTTP/FTP" is checked and applied.
- The impact on the behavior of the FTP/FTPS-server of the check boxes "Enable HTTPS/FTPS" and "Disable HTTP/FTP" is subject of chapter 5.4
- A click on the Delete certificate button causes a pop up window to appear, explaining the consequences



5.2.6.2.5 Events Window

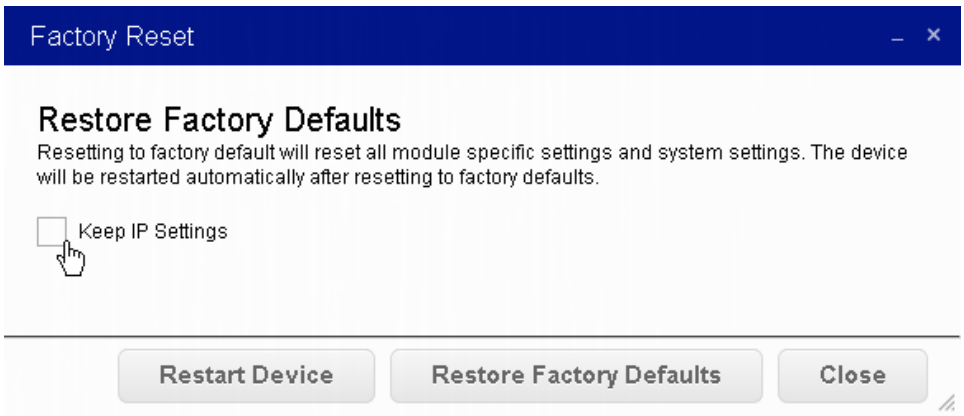


With the Events Window, the SNMP .errorEnableMask (please refer to chapter 5.3.4.4.2) can be accessed via Web Interface.
The different error classes can be enabled or disabled via the related check boxes; by default, all error classes are enabled. If an error class is disabled, an appearing or disappearing event of this error class will not appear/disappear in the System Health Window (chapter 5.2.7). The event and its status (pending or not) are still stated in the Events Tab of the Logs Window (chapter 5.2.5), where the event can be acknowledged and deleted (if not pending anymore).
Note that not all error classes do apply for this implementation.

5.2.6.3 Maintenance

The "**Maintenance**" section in the Setup Window provides access to the Factory Reset Window and to the Firmware Update Window.

5.2.6.3.1 Factory Reset Window



- For just restarting the DEV 2x4x/DEV 2x5x, click on the Restart Device button.

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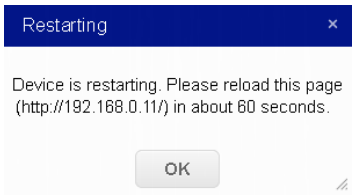
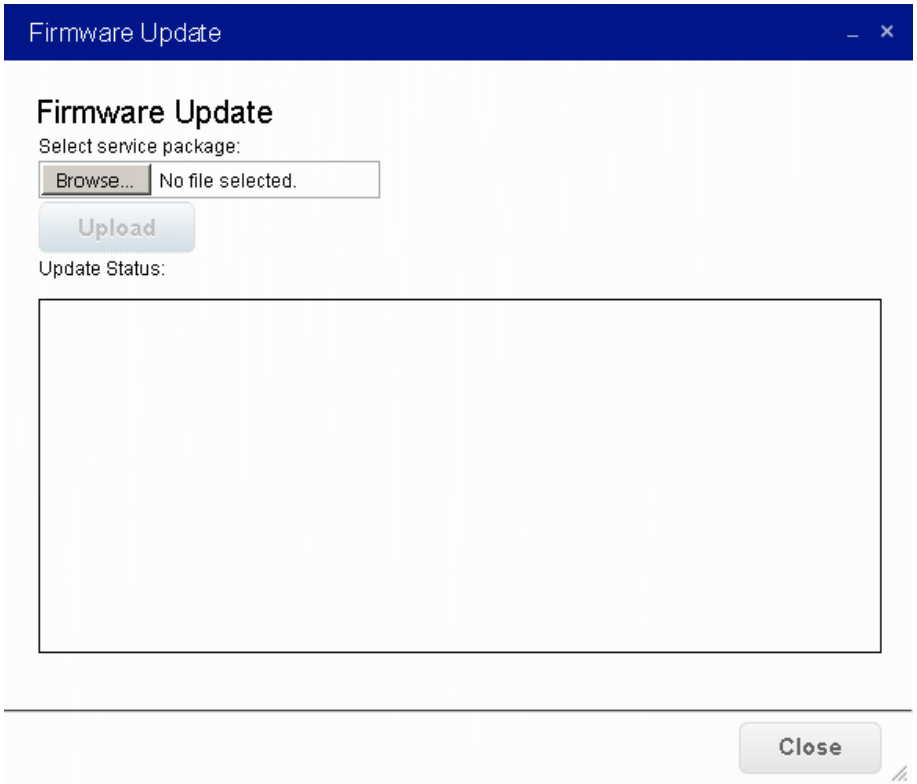
- With a click on the Restore Factory Defaults button, the settings of the equipment are reset to the factory values as stated in chapter 4.3 including all changes on the Web Interface. To maintain the network setup, activate the check box "Keep IP Settings" before operating the Restore Factory Defaults button.

After a click on Restart Device or on Restore Factory Defaults, a reset is initiated indicated via a pop up window



- **Note:**
- If a (the) redundant amplifier path of a DEV 2x5x becomes activated, only a reset to the factory values activates the main amplifier path again.
  Please contact DEV Systemtechnik for assistance if the redundant amplifier path becomes activated once again.

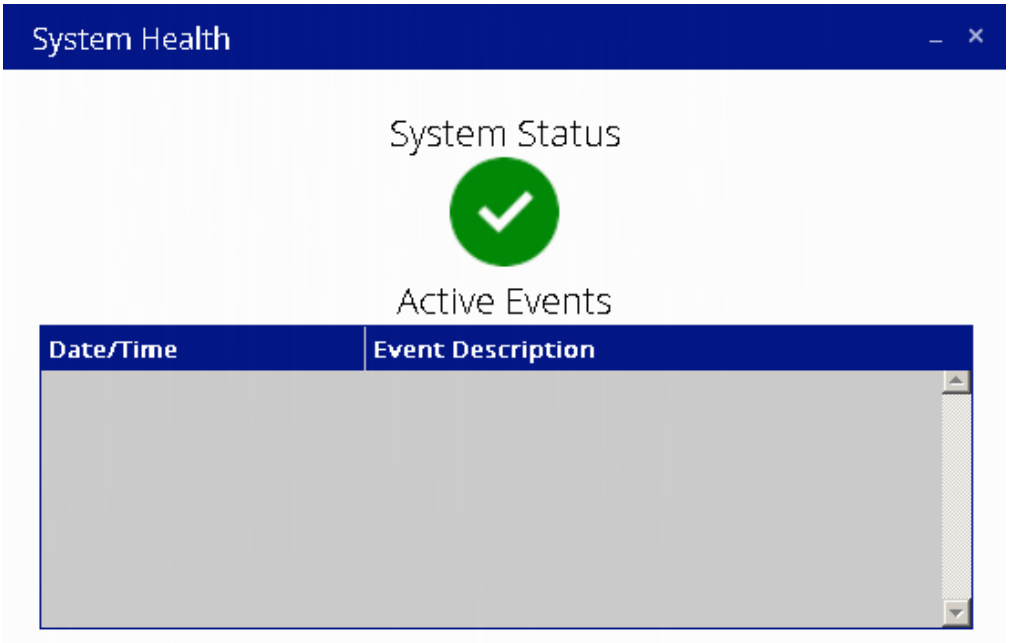
5.2.6.3.2 Firmware Update Window



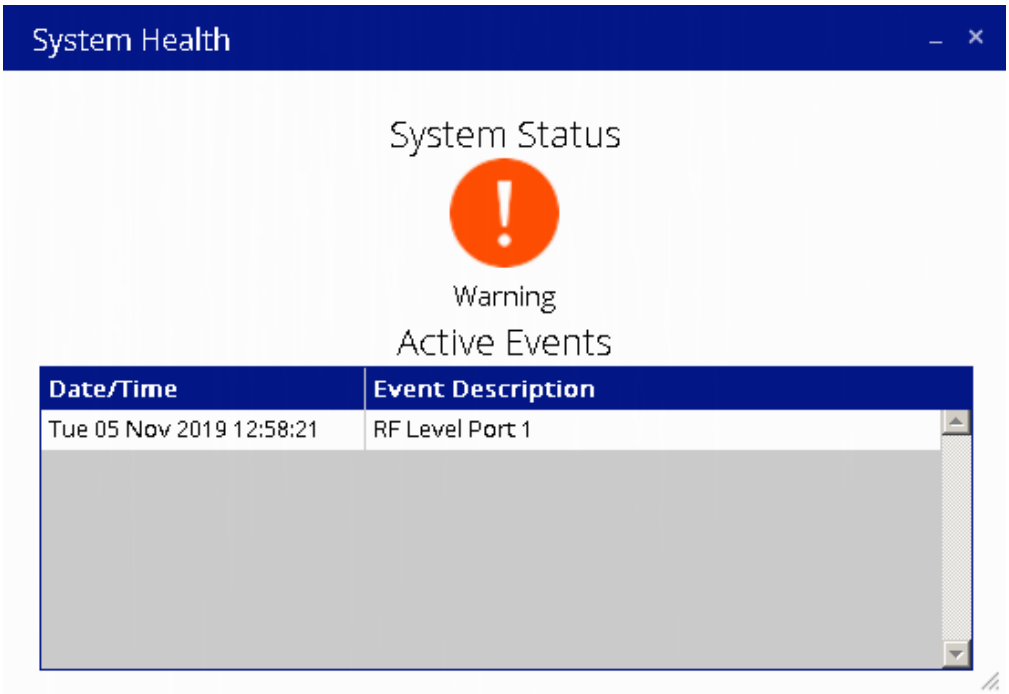
The Firmware Update Window provides means to update firmware, manual, etc. Precondition for an update is a service package provided by DEV Systemtechnik. The input field permits the insertion of a filename including the path where the file resides. Alternatively, the Browse... button can be used to locate the file. After the selection of the file, a click on the Upload button activates the file transfer to the DEV 2x4x/DEV 2x5x. As soon as the upload is finished, the installation is started, please observe the text output, which appears in the "Update Status:" text window.

5.2.7 System Health Window

The System Health Window indicates the overall "**System Status**" and lists pending and not acknowledged events. Without acknowledged events, the "**Active Events**" table is empty, and the "**System Status**" is good ( )



If there are pending and not acknowledged events, the "**System Status**" is "Warning" ( ) and the events are listed in the "**Active Events**" table with a "Date/Time" stamp via the related "Event Description".



To obtain more details on the events and for acknowledging events open the "Event Log" tab in the Logs Window (please refer to chapter 5.2.5).

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5.2.8 About Window

The About Window provides configuration information of the DEV 2x4x/DEV 2x5x. This information is mainly intended for support purposes; part of this information may be requested when contacting DEV Systemtechnik.

Below the header line indicating the equipment type (here "DEV 2243") the following configuration information is listed

- the hardware "MAC Address", here "02-01-33-42-EF-B9"
- the "Firmware Version", here "32-0082 K.01"
- the "OS Version", here "4.19.87H"
- the current "IP Address", here "192.168.0.11"

Additionally, links are provided for:

- downloading (from the equipment) the electronic version of the user manual
- the support email address
- the internet home page

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5.3 Control via Remote Interface: SNMP

5.3.1 SNMP Overview

The "Simple Network Management Protocol" (SNMP) essentially is a request-reply protocol running over UDP (usually ports *161* and *162*). SNMP is an asymmetric protocol operating between two types of management units: managers and agents. A manager is called "Network Management Station" (NMS), which is usually a computer on which one or more management systems are running. On small NMS, command line tools can be used. On larger NMS normally third-party software, also called a "network management suite" collects the data and provides a graphical user interface.
An agent or SNMP-agent is software, which runs on a "Network Element" (NE). NEs in a SNMP environment are called "nodes". An agent collects and stores information from the system, translates the information in SNMP-compatible form, and makes this information available to the manager. The agents manage the "objects" of the node. Each object has an own individual management "Object ID" (OID).
In the system, the objects are the variables, which can be simple values or tables. The object information is stored in the "Management Information Base" (MIB).

Note:

- The iReasoning MIB Browser Personal Edition (including Trap Receiver) by iReasoning Inc. is recommended to test the communication via SNMP.

5.3.2 Private Enterprise Number, Device Index Number

The Private Enterprise Number of DEV Systemtechnik GmbH is 21035.
This MIB is designed in SNMP v2 compliant format starting with the object sections .sysDevModules and .devSysMibConformance, which are not described in this context.

The actual MIB content for some DEV Systemtechnik equipment starts with the object section .devSwitch, which has the Device Index Number 41.

This concatenates to the string i.e., the numerical OID: .1.3.6.1.4.1.21035.41

```
iso.org.dod.internet.private.enterprises.devSystemtechnik.devSwitch
 1 . 3 . 6 .   1    .   4   .     1     .      21035     .    41
```

Note:

- The intention for this MIB design was to keep the structure universal that it can be applied to all kind of DEV Systemtechnik equipment. This implies that only a subset of the MIB functionality is applicable for the DEV 2x4x/DEV 2x5x

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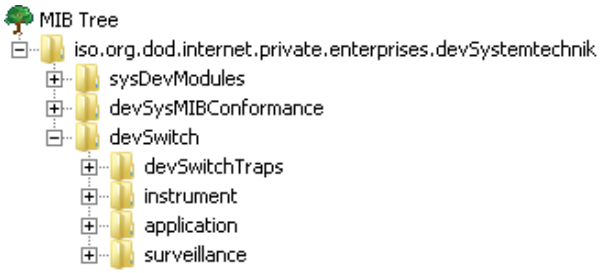
5.3.3 The Switch MIB Tree

The MIB is stored on the equipment and can be downloaded via the SNMP Window of the Web Interface (please refer to chapter 5.2.6.2.3) or directly with a web browser via the link:

`http://<IP-of-the-equipment>/dev-switch.mib`



As shown above, the section .devSwitch is divided in four branches:

- .devSwitchTraps (chapter 5.3.4.1)
- .instrument (chapter 5.3.4.2)
- .application (chapter 5.3.4.3)
- .surveillance (chapter 5.3.4.4)

In the corresponding sub-chapter, the content of each branch is shown expanded and the (relevant) objects within a branch are described.

5.3.4 MIB Objects

The complete name of an MIB object results out of the concatenation of the MIB-string (please refer to chapter 5.3.2)

`iso.org.dod.internet.private.enterprises.devSystemtechnik.devSwitch`

and of the corresponding object.

E.g., complete textual OID of the object `.instrument.controllerMac` is ("⌋" and "⌈" are symbolizing that the two consecutive lines are actually one line):

```
iso.org.dod.internet.private.enterprise.devSystemtechnik⌋
⌈.devSwitch.instrument.controllerMac
```

Note:

- Some of the objects are set to a factory default chapter 4.3.3

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5.3.4.1 .devSwitchTraps

Aside from the SNMP standard trap 'Cold Start Trap', this MIB additionally supports two specific traps.

Note:

- By default, all traps are disabled. Each trap can be enabled individually by setting the appropriate bit in the .trapEnableMask (chapter 5.3.4.4.1)
- Traps for specific error classes can be masked out by clearing the appropriate bits in the .errorEnableMask (please refer to chapter 5.3.4.4.2)

5.3.4.1.1 .devSwitchTraps.trapAlarmText

This trap (number 3, OID: .1.3.6.1.4.1.21035.41.0.3) is triggered if a new event was detected. The trap delivers the appropriate event description string (please refer to chapter 6.1) obtained via .errorDescription (chapter 5.3.4.4.6).
To enable the trap, bit 6 (enum: errortext) in the .trapEnableMask is to be set.

5.3.4.1.2 .devSwitchTraps.trapAlarm

This trap (number 4, OID: .1.3.6.1.4.1.21035.41.0.4) is triggered if a new event was detected or if a pending event disappears. The trap delivers .errorNr (chapter 5.3.4.4.4.2) and .errorPending (chapter 5.3.4.4.4.5) for the related entry in the .errorTable.
Bit 5 (enum: error) in the .trapEnableMask is to be set to enable the trap.

5.3.4.1.3 .devSwitchTraps.trapModeChange

This trap is not required for the functionality of the DEV 2x4x/DEV 2x5x, thus the trap (number 5, OID: .1.3.6.1.4.1.21035.41.0.5) is not implemented.

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5.3.4.2 .instrument

5.3.4.2.1 .instrument.controllerMac

OID:	.1.3.6.1.4.1.21035.41.1.1.*0*
Syntax:	OCTET STRING (SIZE(18))
Access:	read-only
Description:	The individual MAC address of the DEV 2x4x/DEV 2x5x
Example:	"*02-01-33-42-EF-B9*"

5.3.4.2.2 .instrument.firmwareRevision

OID:	.1.3.6.1.4.1.21035.41.1.2.*0*
Syntax:	OCTET STRING (SIZE(32))
Access:	read-only
Description:	The firmware revision of the DEV 2x4x/DEV 2x5x
Example:	"*K.01*"

5.3.4.2.3 .instrument.instrumentType … .instrument.antennaControllerType

The remaining objects of the .instrument branch are not required for the functionality of the DEV 2x4x/DEV 2x5x, thus the following objects are not implemented

- .instrument.instrumentType
- .instrument.antennaControllerType

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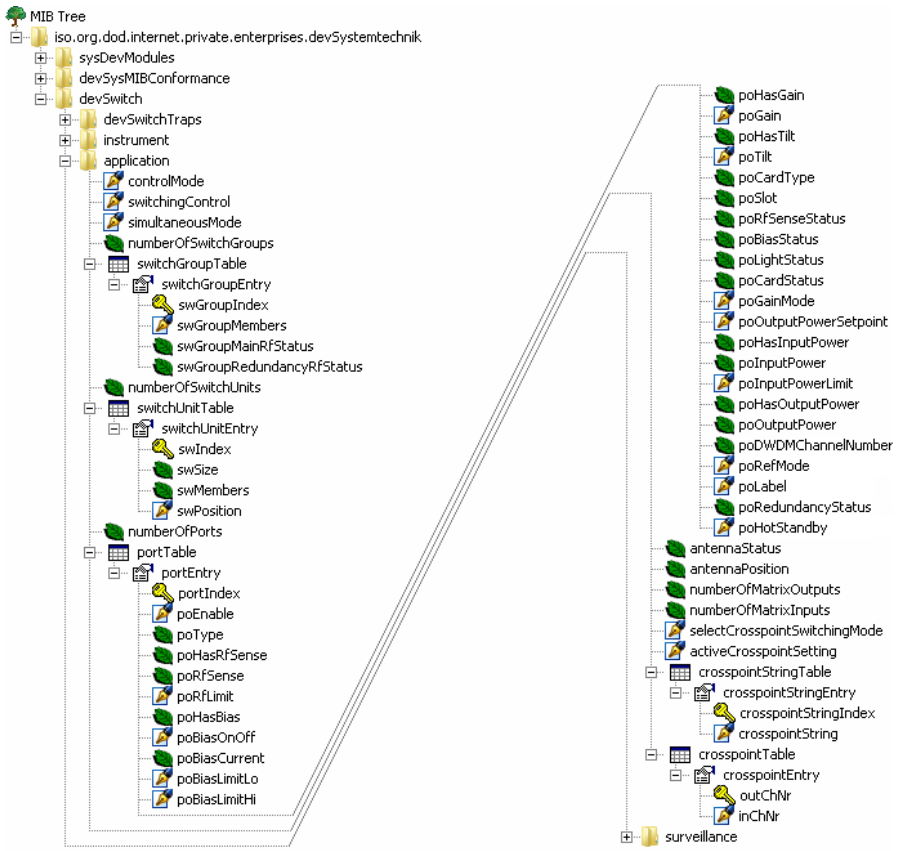
5.3.4.3 .application



5.3.4.3.1 .application.controlMode … .application.switchUnitTable

Some objects of the .application branch are not required for the functionality of the DEV 2x4x/DEV 2x5x thus the following objects are not implemented:

- .application.controlMode
- .application.switchingControl
- .application.simultaneousMode
- .application.numberOfSwitchGroups
- .application.switchGroupTable (including the related table objects)
- .application.numberOfSwitchUnits
- .application.switchUnitTable (including the related table objects)

5.3.4.3.2 .application.numberOfPorts

OID:	.1.3.6.1.4.1.21035.41.2.8.*0*
Syntax:	INTEGER32
Access:	read-only
Description:	The number of externally accessible ports with special properties

Note:

- "*1*" is returned for a DEV 2x4x/DEV 2x5x with a single amplifier installed and "*2*" is returned for a DEV 2x43/DEV 2x53 (with two amplifiers)
- This value determines the maximum index of the following table.

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5.3.4.3.3 .application.portTable

The .portTable describes the special features and settings of each externally accessible port.

5.3.4.3.3.1 .application.portTable.portEntry.portIndex

OID: .1.3.6.1.4.1.21035.41.2.9.1.1.*x*
Syntax: INTEGER32 (1..2048)
Access: not-accessible
Description: A unique value for each table entry ranging between "*1*" and the number of entries (defined by .numberOfPorts, please refer to chapter 5.3.4.3.2)

5.3.4.3.3.2 .application.portTable.portEntry.poEnable

OID: .1.3.6.1.4.1.21035.41.2.9.1.2.*x*
Syntax: INTEGER {disabled (1), enabled (2)}
Access: read-write
Description: Events possibly caused by the port are (can be) enabled or disabled

Note:

- The factory setting is "*2*" (enabled) for all ports.

5.3.4.3.3.3 .application.portTable.portEntry.poType

OID: .1.3.6.1.4.1.21035.41.2.9.1.3.*x*
Syntax: INTEGER32
Access: read-only
Description: The functionality of the port

Note:

- "*0*" (yes) is returned for all ports.

5.3.4.3.3.4 .application.portTable.portEntry.poHasRfSense

OID: .1.3.6.1.4.1.21035.41.2.9.1.4.*x*
Syntax: INTEGER {yes (1), no (2)}
Access: read-only
Description: The indication whether RF level information is provided or not

Note:

- "*1*" (yes) is returned for all DEV 2x4x/DEV 2x5x ports

5.3.4.3.3.5 .application.portTable.portEntry.poRfSense

OID: .1.3.6.1.4.1.21035.41.2.9.1.5.*x*
Syntax: INTEGER32
Access: read-only
Description: The measured RF level in [dBm/10]
Example: "-*129*" ≡ -12.9 dBm

Note:

- "*9999*" means not available or not present.

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5.3.4.3.3.6 .application.portTable.portEntry.poRfLimit

OID: .1.3.6.1.4.1.21035.41.2.9.1.6.*x*
Syntax: INTEGER32
Access: read-write
Description: The RF sensing threshold in [dBm/10] can be read or set
Example: "*-300*" ≡ -30.0 dBm

Note:

- "*9999*" means not available or not present.
- The factory setting is stated in chapter 4.3.1

5.3.4.3.3.7 .application.portTable.portEntry.poHasBias

OID: .1.3.6.1.4.1.21035.41.2.9.1.7.*x*
Syntax: INTEGER {yes (1), no (2)}
Access: read-only
Description: The indication whether LNB power (or bias) is provided or not

Note:

- "*1*" (yes) is returned for (distribution) amplifier ports and
  "*2*" (no) is returned for combining amplifier ports.

5.3.4.3.3.8 .application.portTable.portEntry.poBiasOnOff

OID: .1.3.6.1.4.1.21035.41.2.9.1.8.*x*
Syntax: INTEGER {no (0), on (1), off (2), vertlow (3), horlow (4), verthigh (5), horhigh (6)}
Access: read-write
Description: The LNB power (or bias) supply state can be read or set, possible values

"*0*"	no	for ports with .poHasBias = "*2*" (no)
"*1*"	on	(is not supported)
"*2*"	off	LNB power is (to be) turned off
(*3*…*6*)	LNB power is (to be) turned on and (to be) set to the polarization/band	
"*3*"	vertLow	VL (vertical low: 13 V, 0 Hz)
"*4*"	horLow	HL (horizontal low: 18 V, 0 Hz)
"*5*"	vertHigh	VH (vertical high: 13 V, 22 kHz) or
"*6*"	horHigh	HH (horizontal high: 18 V, 22 kHz)

Note:

- For the LNB power capable (distribution) amplifier ports, the factory setting is "*2*" (off) and can be set to "*3*"…"*6*". For combining amplifier ports "*0*" (no) is returned (and cannot be altered)

5.3.4.3.3.9 .application.portTable.portEntry.poBiasCurrent

OID: .1.3.6.1.4.1.21035.41.2.9.1.9.*x*
Syntax: INTEGER32
Access: read-only
Description: The measured LNB (or bias) current in [mA/10]
Example: "*1850*" ≡ 185.0 mA

Note:

- "*9999*" means not available or not present.

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5.3.4.3.3.10 .application.portTable.portEntry.poBiasLimitLo

OID: .1.3.6.1.4.1.21035.41.2.9.1.10.*x*
Syntax: INTEGER32
Access: read-write
Description: The lower limit of the LNB (or bias) current monitoring interval in [mA/10] can be read or set
Example: "*1000*" ≡ 100.0 mA

Note:

- .poLnbPwrLimitLo can be set for (LNB power capable) ports of a (distribution) amplifier only.
- "*9999*" means not available or not present.
- The factory setting is stated in chapter 4.3.1

5.3.4.3.3.11 .application.portTable.portEntry.poBiasLimitHi

OID: .1.3.6.1.4.1.21035.41.2.9.1.11.*x*
Syntax: INTEGER32
Access: read-write
Description: The upper limit of the LNB (or bias) current monitoring interval in [mA/10] can be read or set
Example: "*2500*" ≡ 250.0 mA

Note:

- .poLnbPwrLimitHi can be set for (LNB power capable) ports of a (distribution) amplifier only.
- "*9999*" means not available or not present.
- The factory setting is stated in chapter 4.3.1

5.3.4.3.3.12 .application.portTable.portEntry.poHasGain

OID: .1.3.6.1.4.1.21035.41.2.9.1.12.*x*
Syntax: INTEGER {yes (1), no (2)}
Access: read-only
Description: The indication whether gain control is provided or not

Note:

- "*1*" (yes) is returned for all DEV 2x4x/DEV 2x5x ports

5.3.4.3.3.13 .application.portTable.portEntry.poGain

OID: .1.3.6.1.4.1.21035.41.2.9.1.13.*x*
Syntax: INTEGER32
Access: read-write
Description: The (manual) gain control value (in tenth) can be read or set
Example: "*50*" ≡ +5.0 dB

Note:

- "*9999*" means not available or not present.
- The factory setting is stated in chapter 4.3.1

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5.3.4.3.3.14 .application.portTable.portEntry.poHasTilt

OID: .1.3.6.1.4.1.21035.41.2.9.1.14.*x*
Syntax: INTEGER {yes (1), no (2)}
Access: read-only
Description: The indication whether slope (or tilt) control is provided or not

Note:

- "*1*" (yes) is returned for all DEV 2x4x/DEV 2x5x ports

5.3.4.3.3.15 .application.portTable.portEntry.poTilt

OID: .1.3.6.1.4.1.21035.41.2.9.1.15.*x*
Syntax: INTEGER32
Access: read-write
Description: The slope (or tilt) control value in [dB/10] can be read or set
Example: "*20*" ≡ +2.0 dB

Note:

- "*9999*" means not available or not present.
- The factory setting is stated in chapter 4.3.1

5.3.4.3.3.16 .application.portTable.portEntry.poCardType

OID: .1.3.6.1.4.1.21035.41.2.9.1.16.*x*
Syntax: OCTET STRING (SIZE(32))
Access: read-only
Description: The DEV number of the module to which the port belongs
Examples: a) "*13-0378*" for a port of a DEV 214x (distribution) amplifier
b) "*13-0386*" for a port of a DEV 224x (combining) amplifier

5.3.4.3.3.17 .application.portTable.portEntry.poSlot

OID: .1.3.6.1.4.1.21035.41.2.9.1.17.*x*
Syntax: OCTET STRING (SIZE(32))
Access: read-only
Description: The slot location of the module to which the port belongs
Examples: a) "*Amp1*" for the first (left) DEV 2x4x/DEV 2x5x amplifier
b) "*Amp2*" for the second (right) DEV 2x43/DEV 2x53 amplifier

5.3.4.3.3.18 .application.portTable.portEntry.poRfSenseStatus

OID: .1.3.6.1.4.1.21035.41.2.9.1.18.*x*
Syntax: INTEGER {ok (1), notok (2), notapplicable (3)}
Access: read-only
Description: The status of the RF level monitoring, possible values
"*1*" ok the RF level is above the RF threshold level
"*2*" not ok the RF level is below the RF threshold level

Note:

- The value "*3*" (not applicable) is not supported

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5.3.4.3.3.19 .application.portTable.portEntry.poBiasStatus

OID: .1.3.6.1.4.1.21035.41.2.9.1.19.*x*
Syntax: INTEGER {ok (1), notok (2), notapplicable (3)}
Access: read-only
Description: The status of the LNB (bias) current monitoring, possible values

"*1*"	ok	LNB power is turned off or the current is within the defined current monitoring interval
"*2*"	not ok	the LNB current exceeds the defined current monitoring interval
"*3*"	not applicable	is returned for all ports of a combining amplifier

5.3.4.3.3.20 .application.portTable.portEntry.poLightStatus

OID: .1.3.6.1.4.1.21035.41.2.9.1.20.*x*
Syntax: INTEGER {ok (1), notok (2), notapplicable (3)}
Access: read-only
Description: The light status

Note:

- "*3*" (not applicable) is returned for all DEV 2x4x/DEV 2x5x ports.

5.3.4.3.3.21 .application.portTable.portEntry.poCardStatus

OID: .1.3.6.1.4.1.21035.41.2.9.1.21.*x*
Syntax: INTEGER {ok (1), notok (2), notapplicable (3)}
Access: read-only
Description: The health status of the card (i.e., amplifier) to which the port belongs, possible values

"*1*"	ok	no events are reported for the amplifier
"*2*"	not ok	an event was detected for the amplifier

Note:

- The value "*3*" (not applicable) is not supported

5.3.4.3.3.22 .application.portTable.portEntry.poGainMode … .poDWDMChannelNumber

Some objects of the .portTable are not required for the functionality of the DEV 2x4x/DEV 2x5x, thus the following objects are not implemented:

- .application.portTable.portEntry.poGainMode
- .application.portTable.portEntry.poOutputPowerSetpoint
- .application.portTable.portEntry.poHasInputPower
- .application.portTable.portEntry.poInputPower
- .application.portTable.portEntry.poInputPowerLimit
- .application.portTable.portEntry.poHasOutputPower
- .application.portTable.portEntry.poOutputPower
- .application.portTable.portEntry.poDWDMChannelNumber

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5.3.4.3.3.23 .application.portTable.portEntry.poRefMode

OID: .1.3.6.1.4.1.21035.41.2.9.1.30.*x*
Syntax: INTEGER {notapplicable (0), off (1), external (2), internal (3)}
Access: read-write
Description: The 10 MHz reference mode

Note:

- "*0*" (not applicable) is returned for all DEV 2x4x/DEV 2x5x ports.

5.3.4.3.3.24 .application.portTable.portEntry.poLabel

OID: .1.3.6.1.4.1.21035.41.2.9.1.31.*x*
Syntax: OCTET STRING
Access: read-write
Description: The port label

Note:

- Alternatively the port label (i.e., the channel name) can be modified in the (related) Amplifier Channel Window of the Web Interface (please refer to chapter 5.2.3.1)

5.3.4.3.3.25 .application.portTable.portEntry.poRedundancyStatus

OID: .1.3.6.1.4.1.21035.41.2.9.1.32.*x*
Syntax: INTEGER {notapplicable (0), main (1), backup (2)}
Access: read-only
Description: The status of the redundant amplifier (DEV 2x5x), possible values

"*0*"	not applicable	is returned for all ports of a DEV 2x4x
"*1*"	main	the main amplifier path is activated
"*2*"	backup	the redundant amplifier path is activated

5.3.4.3.3.26 .application.portTable.portEntry.poHotStandby

OID: .1.3.6.1.4.1.21035.41.2.9.1.33.*x*
Syntax: INTEGER {notapplicable (0), on (1), off (2)}
Access: read-write
Description: The power supply status of the redundant amplifier path (DEV 2x5x) can be read or set, possible values

"*0*"	not applicable	is returned for all ports of a DEV 2x4x (and cannot be altered)
"*1*"	on	the redundant amplifier path is in hot standby
"*2*"	off	the redundant amplifier path is in cold standby

Note:

- Alternatively the power supply of the redundant amplifier path can be modified in the (related) Amplifier Channel Window of the Web Interface (please refer to chapter 5.2.3.1)

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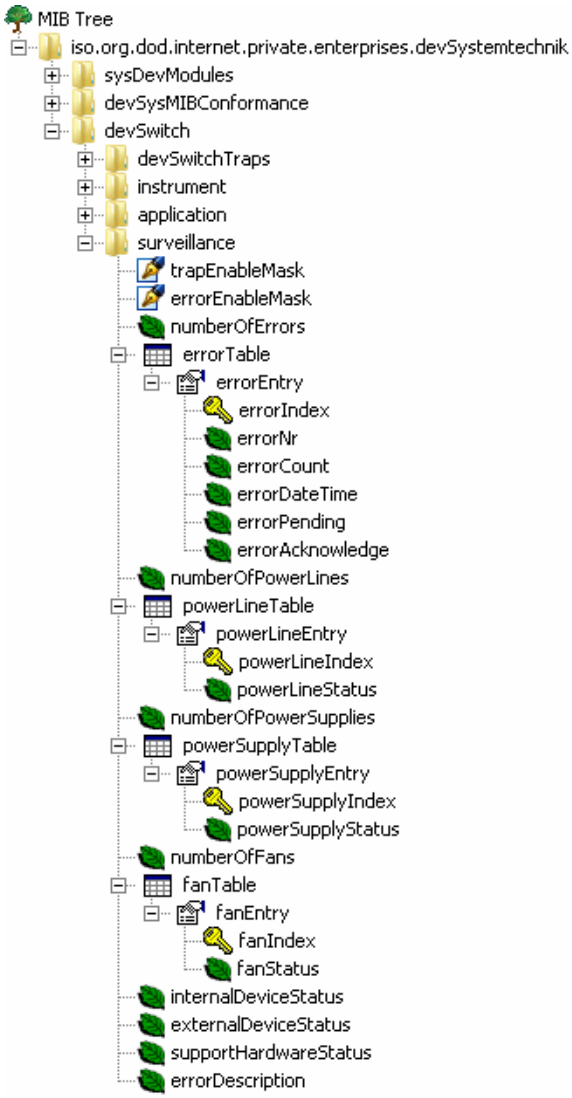
5.3.4.3.4 .application.antennaStatus … .application.crosspointTable

The remaining objects of the .application branch are not required for the functionality of the DEV 2x4x/DEV 2x5x, thus the following objects are not implemented

- .application.antennaStatus
- .application.antennaPosition
- .application.numberOfMatrixOutputs
- .application.numberOfMatrixInputs
- .application.selectCrosspointSwitchingMode
- .application.activeCrosspointSetting
- .application.crosspointStringTable (including the related table objects)
- .application.crosspointTable (including the related table objects)

5.3.4.4 .surveillance



5.3.4.4.1 .surveillance.trapEnableMask

OID: .1.3.6.1.4.1.21035.41.3.1.*0*
Syntax: BITS {unused4 (0), unused3 (1), unused2 (2), unused1 (3), modechange (4), error (5), errortext (6), coldstart (7)}
Access: read-write
Description: Trap enable mask: A trap is enabled if the corresponding bit is set

Note:

- The .trapEnableMask is defined as follows:

Hex	80	40	20	10	08	04	02	01
BITS	0	1	2	3	4	5	6	7
Trap	unused4	unused3	unused2	unused1	modechange	error	errortext	coldstart

  Only bits 5…7 are supported; please refer to chapter 5.3.4.1
- There are differences for the *Set*-method: BITS has the reverse order vs. hex addressing via OCTET STRING e.g., "*0x0F*" enables all supported traps.
- By default, all traps are disabled i.e., all bits are zero.

5.3.4.4.2 .surveillance.errorEnableMask

OID: .1.3.6.1.4.1.21035.41.3.2.*0*
Syntax: INTEGER32 (0..65535)
Access: read-write
Description: Error enable mask; events are grouped into error classes; each error class can be masked out with its corresponding bit set to 0

Note:

- The .errorEnableMask is defined as follows:

Hex	8000	4000	2000	1000	0800	0400	0200	0100	0080	0040	0020	0010	0008	0004	0002	0001
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Class					ANT	COM	LNB	OPT	AMP	MAT	OUT	INP	SWI	FAN	POW	INT

Bit	**Error Class Number**	**Error Class Abbreviation**	**Error Class Name**	**Reference (Chapter)**
0	0	INT	Internal	6.1.3
1	1	POW	Power Supply	6.1.1
2	2	FAN	Fan	(no reference)
3	3	SWI	Switch Module	(no reference)
4	4	INP	Input Module	(no reference)
5	5	OUT	Output Module	(no reference)
6	6	MAT	Matrix Module	(no reference)
7	7	AMP	Amplifier Module	6.1.2
8	8	OPT	Optical Module	(no reference)
9	9	LNB	LNB Power	6.1.2
10	A	COM	Communication	(no reference)
11	B	ANT	Antenna Controller	(no reference)
12…15	C…F		(future use)	(no reference)

- If an error class is masked out, an appearing or disappearing event of this error class is not generating a trap and will not appear/disappear in the System Health Window of the Web Interface (please refer to chapter 5.2.7). The event and its status (pending or not) are still stated in the .errorTable (chapter 5.3.4.4.4) and in the Events Tab of the Logs Window of the Web Interface (chapter 5.2.5), where the event can be acknowledged and deleted (if not pending anymore)
- By default, all events are enabled i.e., the .errorEnableMask is "*65535*".

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5.3.4.4.3 .surveillance.numberOfErrors

OID: .1.3.6.1.4.1.21035.41.3.3.*0*
Syntax: INTEGER32
Access: read-only
Description: The number of events

Note:

- This value is used to determine the maximum index of the following table.

5.3.4.4.4 .surveillance.errorTable

In the entries of the .errorTable, the detected events are recorded classified by .errorNr stating the associated error code. .errorCount in each entry indicates the number of occurrences of the specific event, and .errorDateTime indicates the last occurrence of the event. Furthermore, it is recorded, whether the event is still pending, and whether the event was acknowledged or not. If .numberOfErrors (please refer to chapter 5.3.4.4.3) is "*0*", the .errorTable is not available.

5.3.4.4.4.1 .surveillance.errorTable.errorEntry.errorIndex

OID: .1.3.6.1.4.1.21035.41.3.4.1.1.*x*
Syntax: INTEGER32 (1..512)
Access: not-accessible
Description: A unique value for each table entry ranging between "*1*" and the number of entries (defined by .numberOfErrors, please refer to chapter 5.3.4.4.3)

5.3.4.4.4.2 .surveillance.errorTable.errorEntry.errorNr

OID: .1.3.6.1.4.1.21035.41.3.4.1.2.*x*
Syntax: INTEGER32
Access: read-only
Description: The error code of the event
Example: "*4609*" (for a "Power Line 1" event, please refer to chapter 6.1)

5.3.4.4.4.3 .surveillance.errorTable.errorEntry.errorCount

OID: .1.3.6.1.4.1.21035.41.3.4.1.3.*x*
Syntax: INTEGER32
Access: read-only
Description: The number of occurrences of the event

5.3.4.4.4.4 .surveillance.errorTable.errorEntry.errorDateTime

OID: .1.3.6.1.4.1.21035.41.3.4.1.4.*x*
Syntax: INTEGER32
Access: read-only
Description: The last occurrence of the event
(Unix time stamp, seconds since 01-Jan-1970 00:00:00 UTC)
Example: "*1572966644*" ≡ 05-Nov-2019 15:10:44 CET

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5.3.4.4.4.5 .surveillance.errorTable.errorEntry.errorPending

OID: .1.3.6.1.4.1.21035.41.3.4.1.5.*x*
Syntax: INTEGER {pending (1), gone (2)}
Access: read-only
Description: The last occurrence of the event is pending ("*1*"), or not pending anymore i.e., gone ("*2*")

5.3.4.4.4.6 .surveillance.errorTable.errorEntry.errorAcknowledge

OID: .1.3.6.1.4.1.21035.41.3.4.1.6.*x*
Syntax: INTEGER {notAcknowledged (1), acknowledged (2)}
Access: read-only
Description: The last occurrence of the event is not acknowledged ("*1*"), or acknowledged ("*2*")

5.3.4.4.5 .surveillance.numberOfPowerLines … .surveillance.supportHardwareStatus

The remaining objects of the .surveillance branch are not required for the functionality of the DEV 2x4x/DEV 2x5x, thus the following objects are not implemented

- .surveillance.numberOfPowerLines
- .surveillance.powerLineTable (including the related table objects)
- .surveillance.numberOfPowerSupplies
- .surveillance.powerSupplyTable (including the related table objects)
- .surveillance.numberOfFans
- .surveillance.fanTable (including the related table objects)
- .surveillance.internalDeviceStatus
- .surveillance.externalDeviceStatus
- .surveillance.supportHardwareStatus

5.3.4.4.6 .surveillance.errorDescription

OID: .1.3.6.1.4.1.21035.41.3.14.*0*
Syntax: OCTET STRING (SIZE(64))
Access: accessible-for-notify
Description: .errorDescription contains the latest event description string, which is used for .trapAlarmText; please refer to chapter 5.3.4.1.1

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5.4 FTP/FTPS-Server

The DEV 2x4x/DEV 2x5x provides an integrated FTP/FTPS-server for downloading log files and for uploading firmware updates. Only users with "admin" permission (please refer to chapter 5.2.6.1.1) can access the equipment via FTP/FTPS; all authorization methods (Local, RADIUS, and LDAP; chapter 5.2.6.1.2) are supported. The necessary FTP commands are implemented i.e.,

- LIST and NLST to list the files of the current directory. E.g., `dir` or `ls`
- RETR for downloading the files `error.log` and `user.log`. E.g., `get error.log`
- CWD to change the directory for uploading. E.g., `cd firmware`
- STOR to upload a file (to the `firmware` directory)
  E.g., `put doc_dev214x_224x_215x_225x_91-0154_C.fpk`

Note:

- Active FTP and passive FTP are supported.
- The FTP transfer mode is always binary.
- FTPS can be enabled, and FTP can be disabled via the HTTPS/FTPS Window of the Web Interface (chapter 5.2.6.2.4)
- Explicit FTPS (Port 21 (Control), Port 20 (Data)) and
  implicit FTPS (Port 990 (Control), Port 989 (Data)) are supported.
- If in the HTTPS/FTPS Window (chapter 5.2.6.2.4)…
  …"Enable HTTPS/FTPS" is unchecked:
  The FTP-server listens on Port 21 only. The AUTH TLS command (for explicit TLS) is not accepted
  …"Enable HTTPS/FTPS" is checked and "Disable HTTP/FTP" is unchecked:
  The FTP/FTPS-server listens on Port 21 (unencrypted and via explicit TLS) and on Port 990 (implicit TLS, with PROT C and PROT P supported)
  …"Enable HTTPS/FTPS" is checked and "Disable HTTP/FTP" is checked:
  The FTPS-server listens on Port 990 (implicit TLS) only. Additionally, the data channel (Port 989) has to be encrypted; PROT C is not accepted i.e., the FTPS-client has to send a PROT P to the FTPS-server.

The following lines illustrate a short FTP session (via FTP command line tool):

```
C:\> ftp 192.168.0.11
Connected to 192.168.0.11.
220 DEV Systemtechnik FTP Ready.
Name (192.168.0.11:(none)): ADMIN
331 Please enter your password.
Password: <ADMIN password>
230 Login successful.
ftp> dir
200 PORT command successful.
150 Sending directory listing.
-rwxrwxrwx   1 dev       dev             247 Jan  1  1970 error.log
drwxrwxrwx   1 dev       dev               0 Jan  1  1970 firmware
-rwxrwxrwx   1 dev       dev            1886 Jan  1  1970 user.log
226 Done.
ftp> lcd data
Local directory is C:\data
ftp> get error.log
200 PORT command successful.
150 Sending file contents.
226 Done.
ftp> bye
221 Goodbye
C:\>
```

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6 Event Indication

In this context, an *error* or an *alarm* refer to the indication of an *event*.
The indication of an event is realized via hardware i.e., via LEDs and additionally via software i.e., via Web Interface, via an SNMP trap, or by polling the corresponding SNMP object(s). The hardware indication via LEDs is described in chapter 3.1. The indication via Web Interface is subject of chapters 5.2.5 & 5.2.7; and the indication via SNMP is explained in chapters 5.3.4.1 & 5.3.4.4. The software event indication (i.e., the error codes) is described next.

6.1 Software Event Indication

The DEV 2x4x/DEV 2x5x is capable to detect several types of events that are reported to the Web Interface and to the M&C system. Either the M&C system has to poll the DEV 2x4x/DEV 2x5x for these events, or it is informed via an alarm trap, assuming that the specific SNMP trap is enabled.

6.1.1 Power Events

The power supply is monitored. In case that a missing power line is detected, an appropriate error code is reported.

Error Code		Description
Decimal	**Hexadecimal**	
04609…04610	0x1201…0x1202	Power Line [i] (i = {1…2})

6.1.2 Amplifier Events

The DEV 2x4x/DEV 2x5x provide one or two amplifiers. Different functionalities of the amplifiers are monitored. Event indication is provided for the functionality of the amplifiers, for the RF level functionality, and –if applicable- for the LNB power current monitoring functionality.
If equipped with redundant amplifier (DEV 2x5x), the error codes 0x7E0x indicate which amplifier is switched to the redundant amplifier path.

Error Code		Description
Decimal	**Hexadecimal**	
30976…30977	0x7900…0x7901	Amplifier Module [i] (i = {1…2})
31744…31745	0x7C00…0x7C01	RF Sense Port [i] (i = {1…2})
32256…32257	0x7E00…0x7E01	Redundant Amplifier [i] (i = {1…2})
36864…36865	0x9000…0x9001	LnbPwr Limit Port [i] (i = {1…2})

6.1.3 Other Events

For the sake of completeness, other error codes that might be reported by the DEV 2x4x/DEV 2x5x are listed here. When experiencing one of these events, please contact DEV Systemtechnik for assistance.

Error Code		Description
Decimal	**Hexadecimal**	
2560	0x0A00	Internal
16384	0x4000	System Setup/Config

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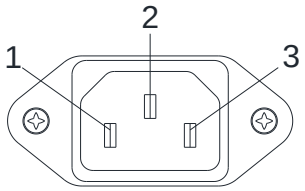
7 Connectors

AC Power Plug

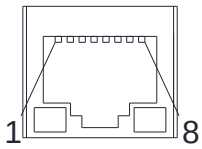
Connector IEC rubber connector (male) IEC 60320 C14
Connector screws none



AC Power Plug	
Pin	**Functionality**
Pin 1	L (Live/Phase)
Pin 2	PE (Protective Earth)
Pin 3	N (Neutral)

Ethernet Connector

Connector RJ-45
Connector assembly none

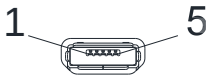


Ethernet Connector	
Pin	**Functionality**
Pin 1	Tx+
Pin 2	Tx-
Pin 3	Rx+
Pins 4…5	do not use
Pin 6	Rx-
Pins 7…8	do not use

- "do not use" means that normally the pin should be left open

USB Connector (@ Front Side Amplifier)

Connector Micro-B USB (receptacle)
Connector assembly none



USB Connector	
Pin	**Functionality**
Pin 1	V_{BUS} (Power,+5 V DC)
Pin 2	D- (Data-)
Pin 3	D+ (Data+)
Pin 4	do not use
Pin 5	GND

- "do not use" means that normally the pin should be left open

8 Technical Data and Options

The technical data and the options are stated in the related spec sheets; please refer to www.dev-systemtechnik.com.

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9 Conformance Declarations

9.1 Certificate of Conformance

Since DEV Systemtechnik develops and produces in Germany, the Certificate of Conformance for each product or product series is to be issued in German (as stated on the next page).

For the ease of comprehension, the harmonized standards that were taken into consideration during development and production are listed here in English:

DIN EN 50083-2:2016-03	Cable networks for television signals, sound signals and interactive services - Part 2: Electromagnetic compatibility for equipment; German version EN 50083-2:2012 + A1:2015
DIN EN 55022:2011-12	Information technology equipment - Radio disturbance characteristics - Limits and methods of measurement (CISPR 22:2008, modified); German version EN 55022:2010
DIN EN 55024:2016-05	Information technology equipment - Immunity characteristics - Limits and methods of measurement (CISPR 24:2010 + Cor.:2011 + A1:2015); German version EN 55024:2010 + A1:2015
DIN EN 61000-3-2:2015-03	Electromagnetic compatibility (EMC) - Part 3-2: Limits - Limits for harmonic current emissions (equipment input current ≤ 16 A per phase) (IEC 61000-3-2:2014); German version EN 61000-3-2:2014
DIN EN 61000-3-3:2014-03	Electromagnetic compatibility (EMC) - Part 3-3: Limits - Limitation of voltage changes, voltage fluctuations, and flicker in public low-voltage supply systems, for equipment with rated current ≤16 A per phase and not subject to conditional connection (IEC 61000-3-3:2013); German version EN 61000-3-3:2013
DIN EN 62368-1:2016-05	Audio/video, information and communication technology equipment - Part 1: Safety requirements (IEC 62368-1:2014, modified + Cor.:2015); German version EN 62368-1:2014 + AC:2015

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Konformitätserklärung nach
EMV-Richtlinie (2014/30/EU)
Niederspannungsrichtlinie (2014/35/EU)

Der Hersteller:	DEV Systemtechnik GmbH Grüner Weg 4A 61169 Friedberg

erklärt hiermit in alleiniger Verantwortung, dass nachfolgend bezeichnete Produkte in denen von uns in Verkehr gebrachten Ausführungen:

Produktbezeichnungen:	Managed/Redundant Distribution/Combining Amplifiers; 800...2450 MHz
Produktnummern:	DEV 2142, DEV 2142/N, DEV 2143, DEV 2145, DEV 2151, DEV 2151/N, DEV 2152, DEV 2152/N, DEV 2153, DEV 2155, DEV 2242, DEV 2242/N, DEV 2243, DEV 2245, DEV 2252, DEV 2252/N, DEV 2253, DEV 2255
Baujahr:	ab 2019

auf die sich diese Erklärung bezieht, mit den folgenden Normen übereinstimmen.

Bei der Konzipierung und beim Bau der Geräte wurden die folgenden harmonisierten Normen berücksichtigt:

DIN EN 50083-2:2016-03	Kabelnetze für Fernsehsignale, Tonsignale und interaktive Dienste - Teil 2: Elektromagnetische Verträglichkeit von Geräten; Deutsche Fassung EN 50083-2:2012 + A1:2015
DIN EN 55022:2011-12	Einrichtungen der Informationstechnik - Funkstöreigenschaften - Grenzwerte und Messverfahren (CISPR 22:2008, modifiziert); Deutsche Fassung EN 55022:2010
DIN EN 55024:2016-05	Einrichtungen der Informationstechnik - Störfestigkeitseigenschaften - Grenzwerte und Prüfverfahren (CISPR 24:2010 + Cor.:2011 + A1:2015); Deutsche Fassung EN 55024:2010 + A1:2015
DIN EN 61000-3-2:2015-03	Elektromagnetische Verträglichkeit (EMV) - Teil 3-2: Grenzwerte - Grenzwerte für Oberschwingungsströme (Geräte-Eingangsstrom ≤16 A je Leiter) (IEC 61000-3-2:2014); Deutsche Fassung EN 61000-3-2:2014
DIN EN 61000-3-3:2014-03	Elektromagnetische Verträglichkeit (EMV) - Teil 3-3: Grenzwerte - Begrenzung von Spannungsänderungen, Spannungsschwankungen und Flicker in öffentlichen Niederspannungs-Versorgungsnetzen für Geräte mit einem Bemessungsstrom ≤16 A je Leiter, die keiner Sonderanschlussbedingung unterliegen (IEC 61000-3-3:2013); Deutsche Fassung EN 61000-3-3:2013
DIN EN 62368-1:2016-05	Einrichtungen für Audio/Video-, Informations- und Kommunikationstechnik - Teil 1: Sicherheitsanforderungen (IEC 62368-1:2014, modifiziert + Cor.:2015); Deutsche Fassung EN 62368-1:2014 + AC:2015

DEV Systemtechnik GmbH
Friedberg, 15.02.2023

Manfred Mettendorff
Managing Director

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9.2 EU Declaration of Conformity

As original manufacturer of Electrical & Electronic Equipment (EEE)

DEV Systemtechnik GmbH
Grüner Weg 4A
61169 Friedberg
GERMANY

declares under sole responsibility that the

Products

DEV 2142	1:8	Managed Distribution Amplifier; 800...2450 MHz
DEV 2142/N	1:8	Managed Distribution Amplifier; 800...2450 MHz; 50 Ohm, N (f)
DEV 2143	2*1:8	Managed Distribution Amplifier; 800...2450 MHz
DEV 2145	1:16	Managed Distribution Amplifier; 800...2450 MHz
DEV 2151		Redundant Amplifier; 800...2450 MHz
DEV 2151/N		Redundant Amplifier; 800...2450 MHz; 50 Ohm, N (f)
DEV 2152	1:8	Redundant Distribution Amplifier; 800...2450 MHz
DEV 2152/N	1:8	Redundant Distribution Amplifier; 800...2450 MHz; 50 Ohm, N (f)
DEV 2153	2*1:8	Redundant Distribution Amplifier; 800...2450 MHz
DEV 2155	1:16	Redundant Distribution Amplifier; 800...2450 MHz
DEV 2242	8:1	Managed Combining Amplifier; 800...2450 MHz
DEV 2242/N	8:1	Managed Combining Amplifier; 800...2450 MHz; 50 Ohm, N (f)
DEV 2243	2*8:1	Managed Combining Amplifier; 800...2450 MHz
DEV 2245	16:1	Managed Combining Amplifier; 800...2450 MHz
DEV 2252	8:1	Redundant Combining Amplifier; 800...2450 MHz
DEV 2252/N	8:1	Redundant Combining Amplifier; 800...2450 MHz; 50 Ohm, N (f)
DEV 2253	2*8:1	Redundant Combining Amplifier; 800...2450 MHz
DEV 2255	16:1	Redundant Combining Amplifier; 800...2450 MHz

Options for Amplifiers (DEV 2151, DEV 2151/N)

Option 29	Additional Amplifier

Options for Amplifiers (DEV 2151)

Option vAin50 (vAout50)	One Amplifier Input (Output) Port 50 Ohm, SMA (f)
Option vAin50B (vAout50B)	One Amplifier Input (Output) Port 50 Ohm, BNC (f)
Option vAin75 (vAout75)	One Amplifier Input (Output) Port 75 Ohm, F (f)
Option vAin75B (vAout75B)	One Amplifier Input (Output) Port 75 Ohm, BNC (f)

Options for Distribution Amplifiers (DEV 2142, DEV 2143, DEV 2145, DEV 2152, DEV 2153, DEV 2155)

Option vSin50 (vSout50)	One (Eight) Splitter Input (Output) Port(s) 50 Ohm, SMA (f)
Option vSin50B (vSout50B)	One (Eight) Splitter Input (Output) Port(s) 50 Ohm, BNC (f)
Option vSin75 (vSout75)	One (Eight) Splitter Input (Output) Port(s) 75 Ohm, F (f)
Option vSin75B (vSout75B)	One (Eight) Splitter Input (Output) Port(s) 75 Ohm, BNC (f)

Options for Combining Amplifiers (DEV 2242, DEV 2243, DEV 2245, DEV 2252, DEV 2253, DEV 2255)

Option vCin50 (vCout50)	Eight (One) Combiner Input (Output) Ports (Port) 50 Ohm, SMA (f)
Option vCin50B (vCout50B)	Eight (One) Combiner Input (Output) Ports (Port) 50 Ohm, BNC (f)
Option vCin75 (vCout75)	Eight (One) Combiner Input (Output) Ports (Port) 75 Ohm, F (f)
Option vCin75B (vCout75B)	Eight (One) Combiner Input (Output) Ports (Port) 75 Ohm, BNC (f)

to which this declaration refers to, is in conformity with Directive 2011/65/EU of the European Parliament and of the Council of 8-Jun-2011 on the restriction of the use of certain hazardous substances in electrical and electronic equipment (RoHS).

Signed for and on behalf of DEV Systemtechnik GmbH

Name: Manfred Mettendorff

Position: Managing Director

Place and date of issue: Friedberg, 15-Feb-2023

Signature:

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10 Glossary

AC	**A**lternating **C**urrent
CEST	**C**entral **E**uropean **S**ummer **T**ime
CET	**C**entral **E**uropean **T**ime
DA	**D**istribution **A**mplifier
DC	**D**irect **C**urrent
DHCP	**D**ynamic **H**ost **C**onfiguration **P**rotocol
FTP	**F**ile **T**ransfer **P**rotocol: Standard network protocol that is used to transfer files between a client and a server
FTPS	FTP **S**ecure: Extension to → FTP, adding support for the → TLS and the → SSL cryptographic protocols
HH	**H**orizontal **H**igh: Polarization/Band for LNB powering (18 V, 22 kHz) (see also → HL, → VL, → VH)
HL	**H**orizontal **L**ow: Polarization/Band for LNB powering (18 V, 0 Hz) (see also → HH, → VL, → VH)
HTTP	**H**yper**T**ext **T**ransfer **P**rotocol: Standard network protocol for hypermedia information
HTTPS	HTTP **S**ecure: Extension to → HTTP, adding support for the → TLS and the → SSL cryptographic protocols
IP	**I**nternet **P**rotocol
LDAP	**L**ightweight **D**irectory **A**ccess **P**rotocol: Open, industry standard network protocol for distributed directory information services → RADIUS
LED	**L**ight **E**mitting **D**iode
LNB	**L**ow **N**oise **B**lock Converter
M&C System	**M**anagement and **C**ontrol System
MAC	**M**edia **A**ccess **C**ontrol (address) is the hardware address for the unique identification of equipment within a network
MIB	**M**anagement **I**nformation **B**ase
NE	**N**etwork **E**lement
NMS	**N**etwork **M**anagement **S**ystem
NTP	**N**etwork **T**ime **P**rotocol
OID	**O**bject **ID**entifier
PC	**P**ersonal **C**omputer
PE	**P**rotective **E**arth
PEM	**P**rivacy-enhanced **E**lectronic **M**ail: .pem is the file format for certificates
RADIUS	**R**emote **A**uthentication **D**ial-**I**n **U**ser **S**ervice: Network protocol for centralized authentication, authorization, and accounting → LDAP
RF	**R**adio **F**requency
RU	**R**ack **U**nit: Units of vertical height. 1 RU = 1 ¾" = 44.45 mm Sometimes simply referred to as "U" (Unit)
SNMP	**S**imple **N**etwork **M**anagement **P**rotocol
SSL	**S**ecure **S**ockets **L**ayer: Cryptographic protocol → FTPS → HTTPS
TCP	**T**ransmission **C**ontrol **P**rotocol
TCP/IP	TCP**/I**nternet **P**rotocol
TLS	**T**ransport **L**ayer **S**ecurity: Cryptographic protocol → FTPS → HTTPS
UDP	**U**ser **D**atagram **P**rotocol
URL	**U**niform **R**esource **L**ocator
UTC	**U**niversal **T**ime **C**oordinated
VH	**V**ertical **H**igh: Polarization/Band for LNB powering (13 V, 22 kHz) (see also → HH, → HL, → VL)
VL	**V**ertical **L**ow: Polarization/Band for LNB powering (13 V, 0 Hz) (see also → HH, → HL, → VH)

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11 Notes

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